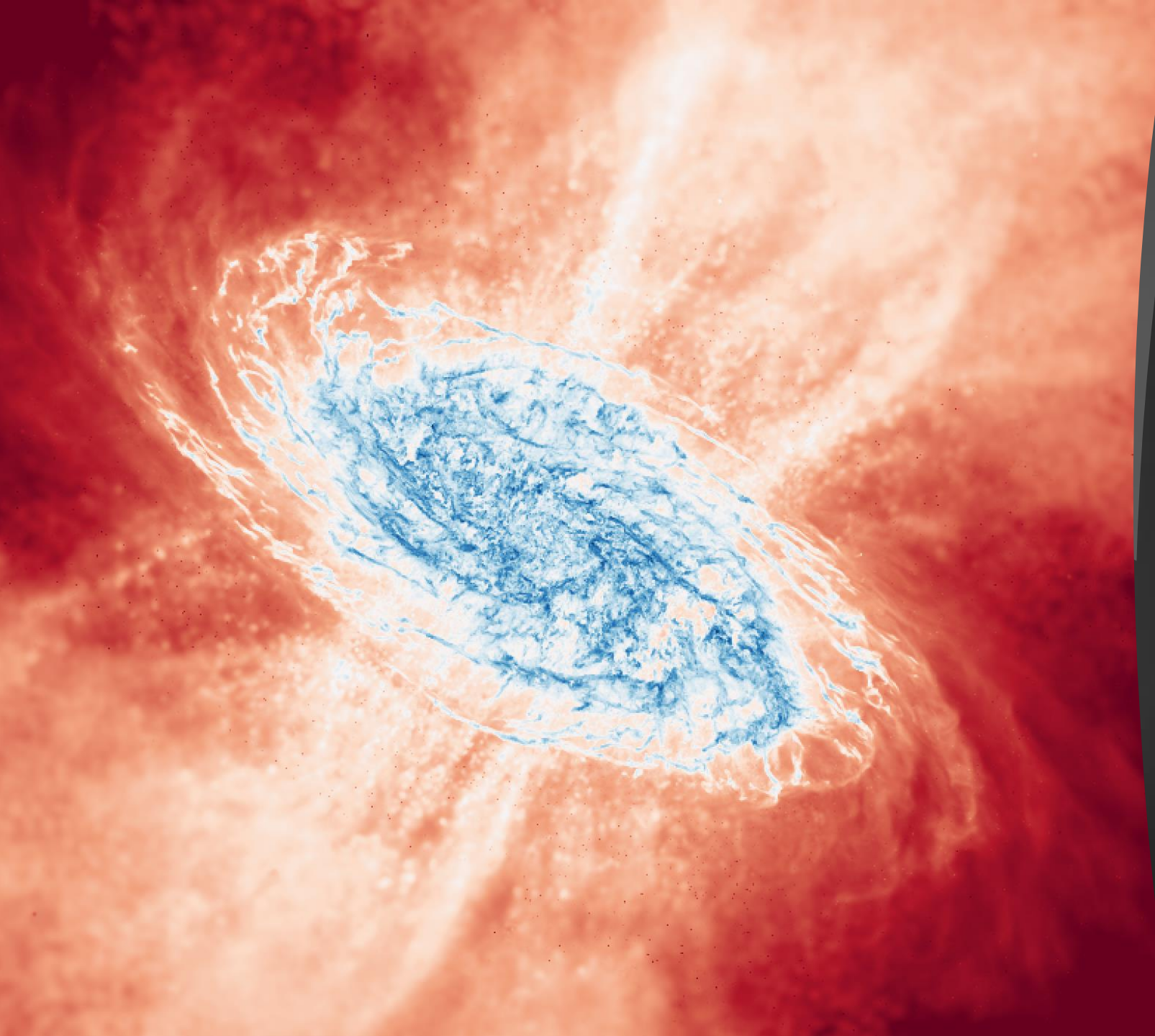


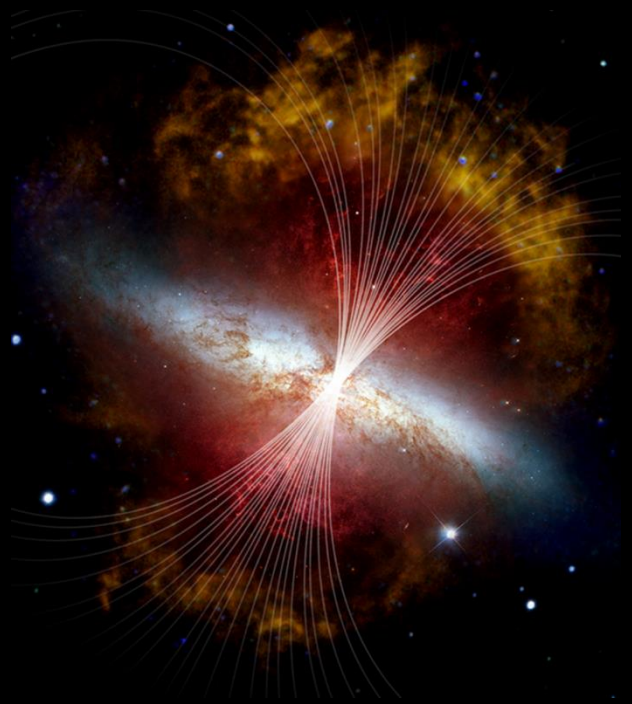
Simulating Galactic Dynamo Processes with Smoothed Particle Magnetohydrodynamics

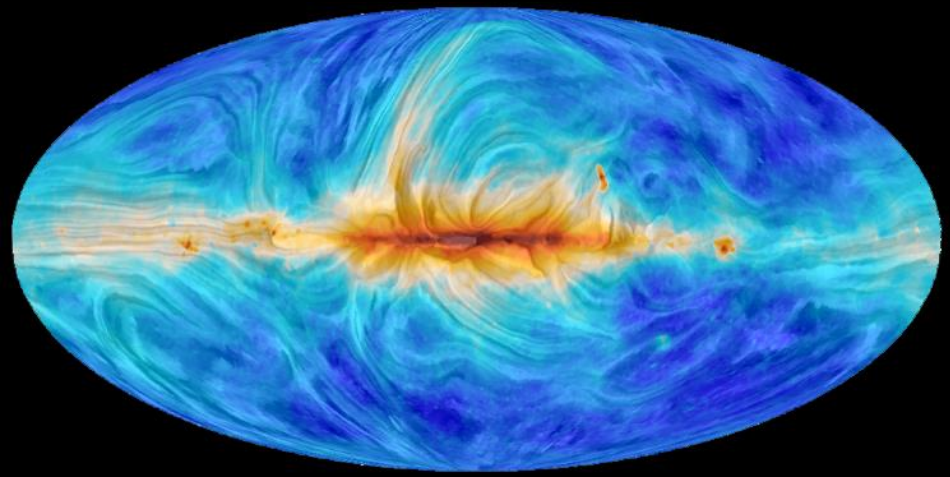
Robert Wissing

Institute of Theoretical Astrophysics, University of Oslo

Magnetic fields in galaxies







Planck Satellite
Milky Way (Planck Collaboration 2016)

- Ubiqitious in our Universe: the Earth, the Sun, in star forming region (GMCs), galaxies and the CGM/IGM;
- Essential component in star formation;
- Can reach equipartition with thermal and turbulent energy in galaxies -> dynamically important (e.g., Beck+2015);
- Transport of cosmic rays, potentially driving galactic outflows (e.g., Girichidis+2016a);

Origins and amplification of the B-fields

- Primordial seed B fields are *weak*: *e.g.* 10^{-20} G from Bierman battery
- Stellar winds, SNe, AGN feedback inject additional B fields;
- Equipartition -> ~ μG in the MW- a few 100 μG in starburst galaxies (e.g., M82)
- B fields are amplified already in high-z galaxies;
- Galactic dynamo with exponential growth required;
  - Mean-field dynamo
  - Small-scale dynamo

Mean-field Dynamo

- Generate large-scale, ordered B field
  At scale > turbulence driving

mean field

$B = \bar{B} + b, \qquad U = \bar{U} + u$ fluctuation

$$\frac{\partial \overline{B}}{\partial t} = \nabla \times (\overline{U} \times \overline{B}) + \nabla \times \mathcal{E} + \eta \nabla^2 \overline{B}$$

$\mathcal{E} = \overline{u \times b}$ electromotive force (from small-scale motions)

$$\mathcal{E}_i = \alpha_{ij} \overline{B_j} - \eta_{ij} \overline{J_j} + \gamma_{ij} \overline{\Omega_j} + \ldots$$

Mean-field Dynamo

- Generate large-scale, ordered B field
  At scale > turbulence driving
- $\alpha\omega$ dynamo (Parker+1955): shearing motion (ω-effect) and helicities of the velocity fields (α-effect);

mean field

$B = \bar{B} + b, \qquad U = \bar{U} + u$ fluctuation

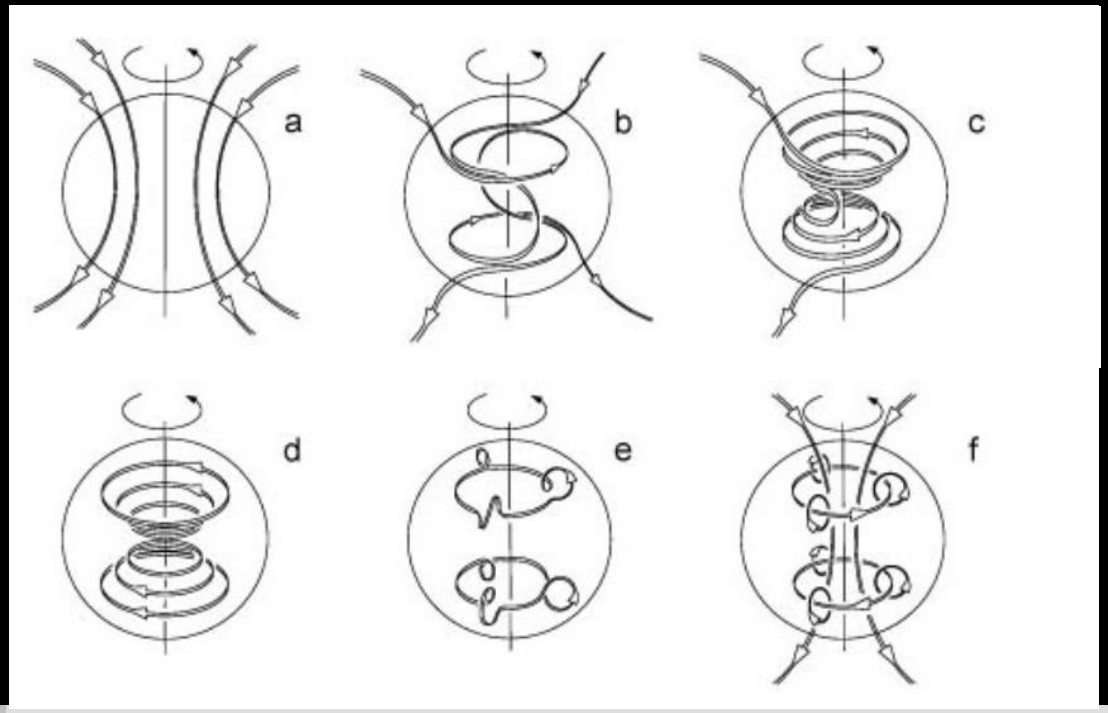
$$\frac{\partial \overline{B}}{\partial t} = \nabla \times (\overline{U} \times \overline{B}) + \nabla \times \mathcal{E} + \eta \nabla^2 \overline{B}$$

$\mathcal{E} = \overline{u \times b}$ electromotive force (from small-scale motions)

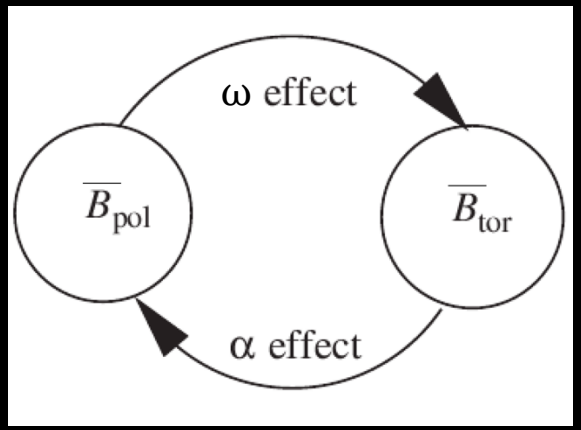
α effect

$$\mathcal{E}_i = \alpha_{ij} \overline{B_j} - \eta_{ij} \overline{J_j} + \gamma_{ij} \overline{\Omega_j} + \ldots$$

ω-effect
$\bar{U} \propto \omega$



$\alpha\omega$ dynamo



Mean-field Dynamo

mean field

$B = \bar{B} + b, \qquad U = \bar{U} + u$ fluctuation

$$\frac{\partial \overline{B}}{\partial t} = \nabla \times (\overline{U} \times \overline{B}) + \nabla \times \mathcal{E} + \eta \nabla^2 \overline{B}$$

- Generate large-scale, ordered B field
  At scale > turbulence driving
- $\alpha\omega$ dynamo (Parker+1955): shearing motion (ω-effect) and helicities of the velocity fields (α-effect);
- Other mean-field dynamo rely on a similar process

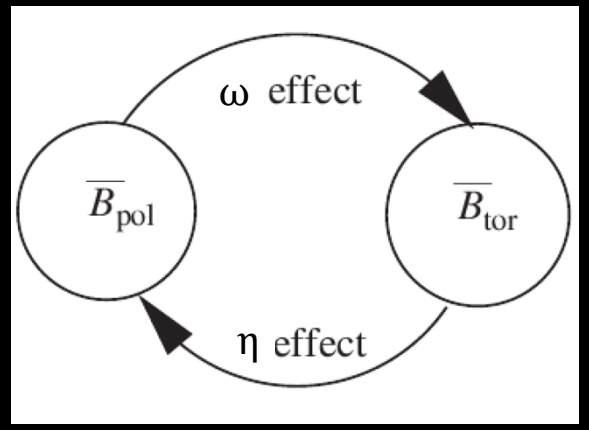
$\mathcal{E} = \overline{u \times b}$ electromotive force (from small-scale motions)

α effect

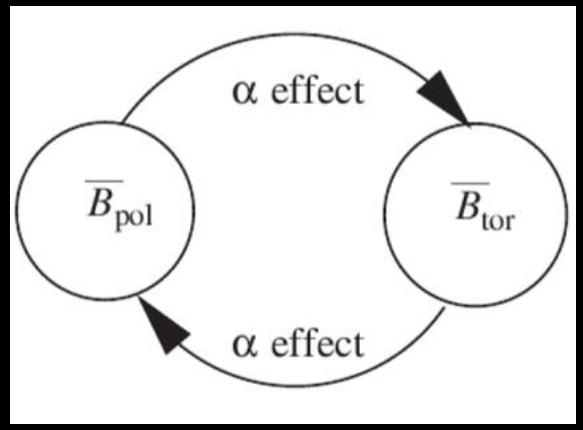
$$\mathcal{E}_i = \alpha_{ij} \overline{B_j} - \eta_{ij} \overline{J_j} + \gamma_{ij} \overline{\Omega_j} + \ldots$$

ω-effect
$\bar{U} \propto \omega$

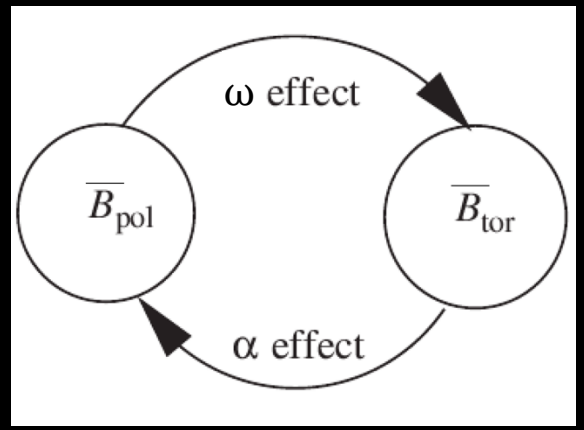
Shear-current dynamo



α^2 dynamo

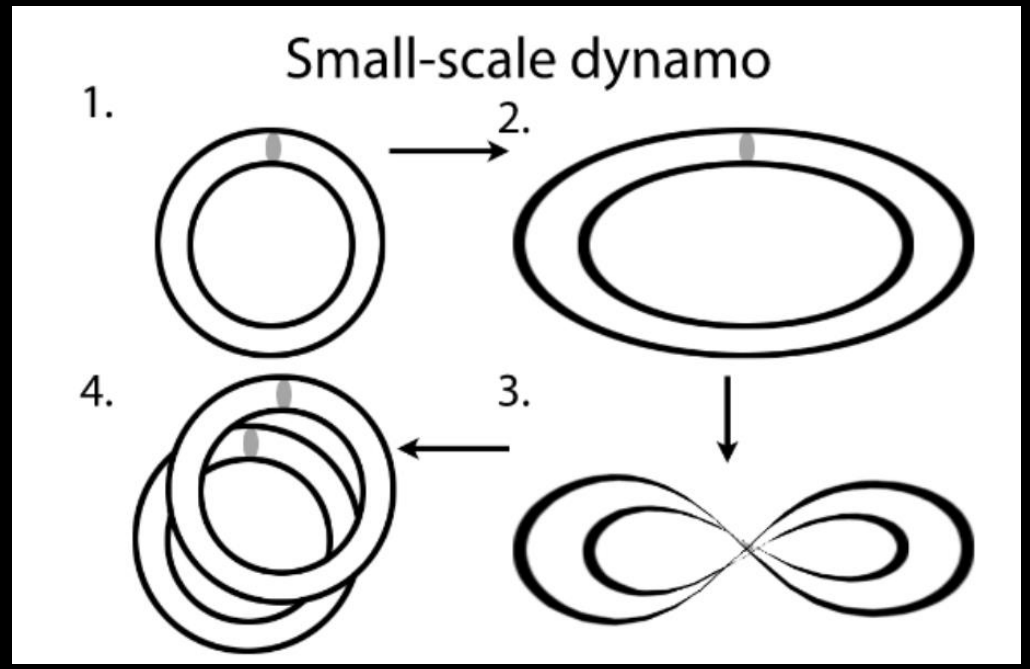


$\alpha\omega$ dynamo



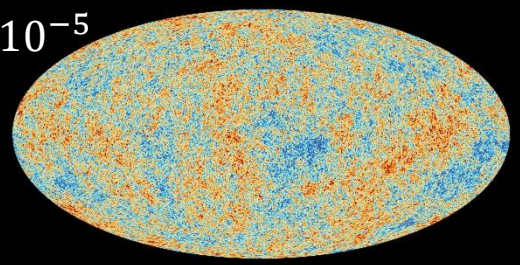
Small-scale Dynamo

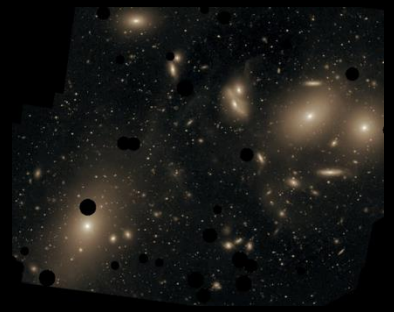
- Streching, twisting and folding of field lines by turbulent flows
- Create randomly oriented B-field
  At scale < turbulence driving
- e-folding time: viscous scale eddies (smaller scale amplies faster); 10-100 Myr in galaxies (Schekochihin+2004)
- **Solenoidal** (rotational) mode turbulence is essential to excite the dynamo;



Galaxy Formation

: Highly non-linear process
: All scales are connected





Virialized halos $\delta \sim 10^2$
$L \sim 100kpc - Mpc$



Galaxies $L \sim 1 - 10kpc$ $\delta \sim 10^5 - 10^6$
Typical ISM: $n \sim 1cm^{-3}$

Turbulence drivers(CGM & ISM)
Gas accretion
Gravitational instability
Shearing of spiral arms

"Feedback":
Radiative
Mechanical
Chemical
Non-thermal





Supermassive black holes
Accretion disk $n \sim 10^7 cm^{-3}$
$L < 1pc$



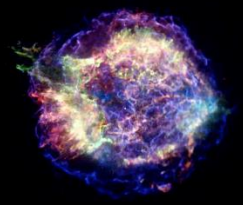
Turbulence drivers (ISM):
SNe, stellar winds, AGN jets

Molecular clouds
$n \sim 10^2 cm^{-3}, L \sim 1 - 100pc$

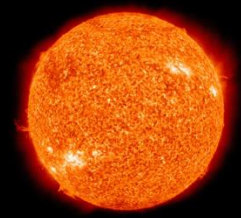
Galactic winds:
Field advection into the CGM
Inject turbulence in the CGM
Fountain motion near disc



Galactic winds:
$L \sim 100kpc - 1Mpc$



SN ejecta:
$L \sim 1 - 50\,pc$



Stars: $n \sim 10^{24} cm^{-3}$

Goals of this thesis work

- Develop a numerical scheme for galaxy simulations that can properly model magnetic fields
- Application of the method to challenging physical processes:

Magnetized core collapse and jet formation

The magneto-rotational instability

- Explore what kind of dynamo processes are active during galaxy formation

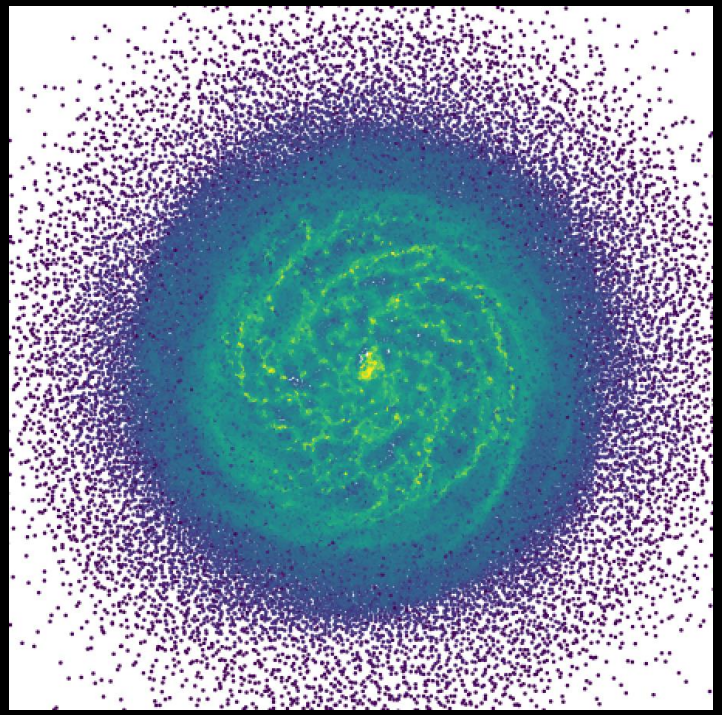
How does galactic dynamo depend on global disc dynamics (e.g., sprial arms) and local turbulence drivers like SN feedback?

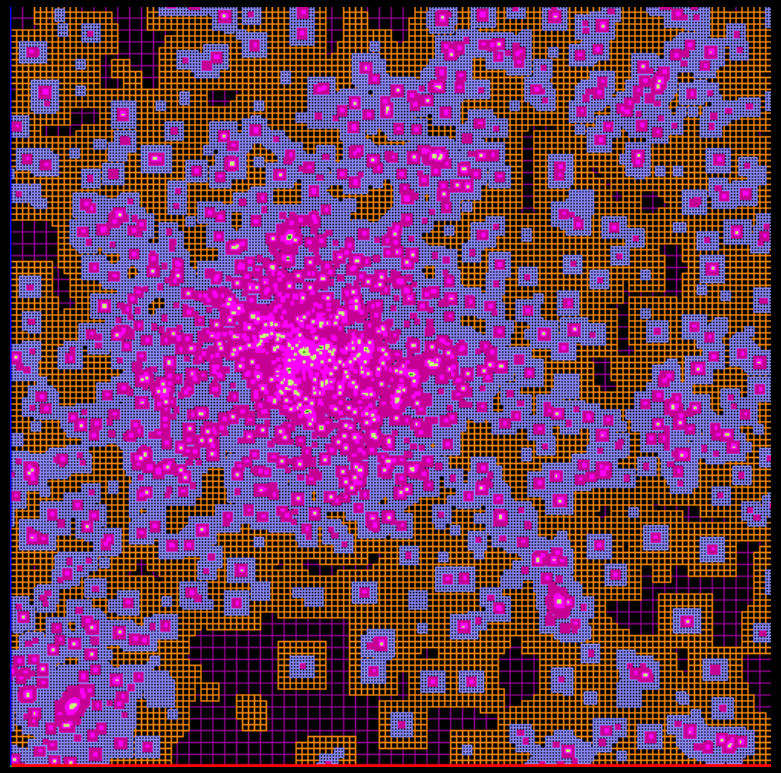
What is the role of galactic outflows and galactic fountains?

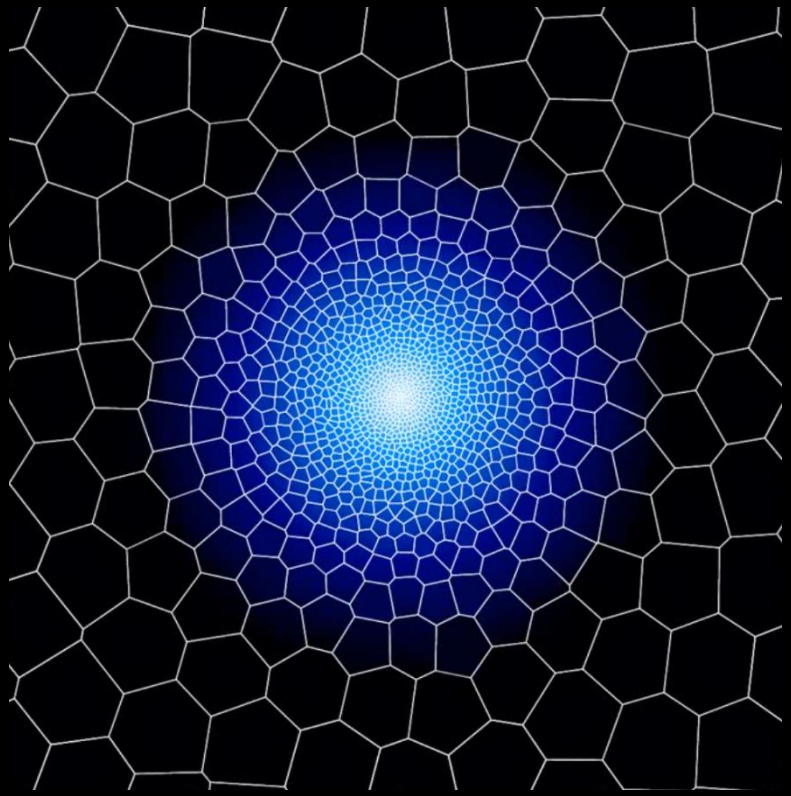
How does the amplification depend on fluid parameters (dissipation parameters, Reynolds/magnetic Reynolds number), and resolution?

Numerical method

- Smoothed particle hydrodynamics (SPH)
- AMR grid
- Moving mesh
- Meshless finite mass (MFM)

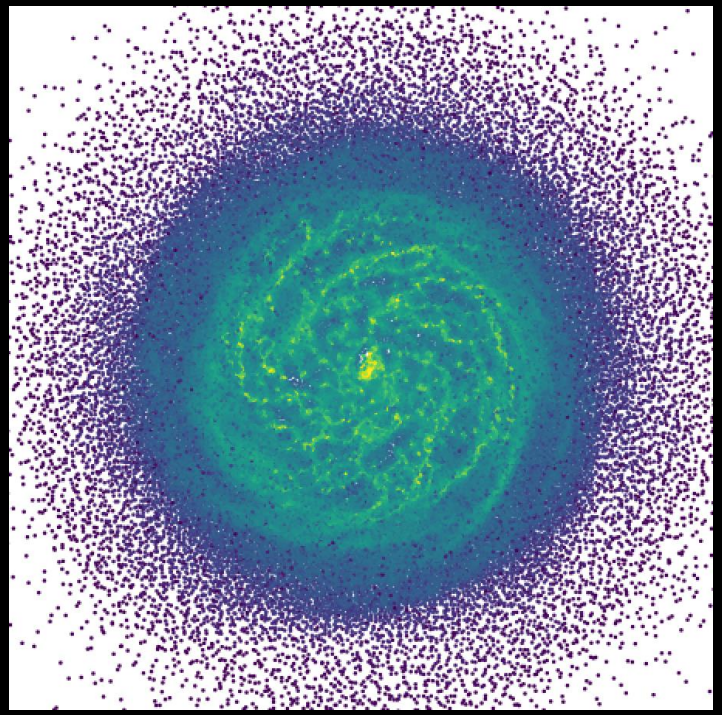






Numerical method

- Smoothed particle hydrodynamics (SPH)

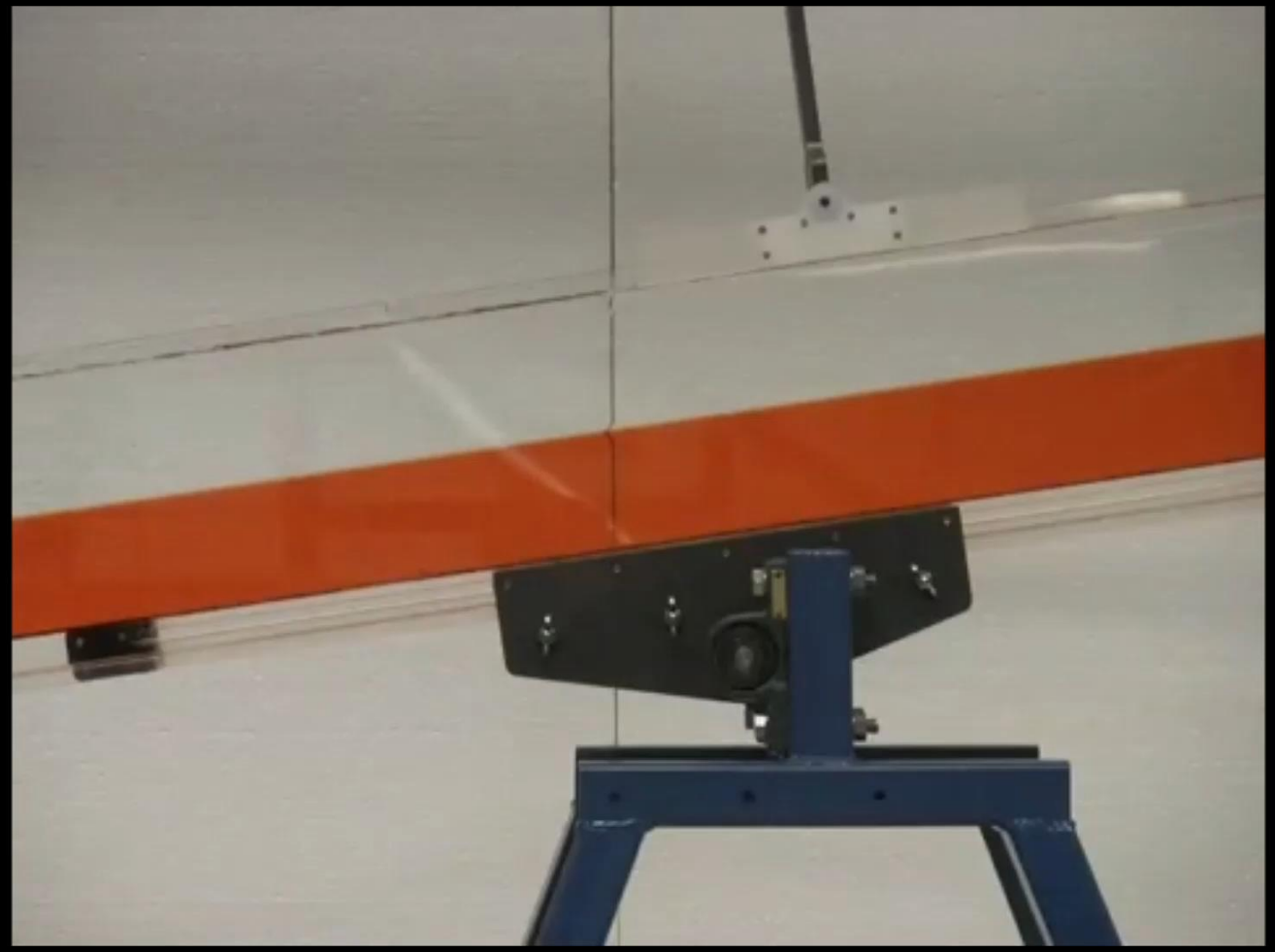


Advantages of SPH for astrophysics:

1. Natively adaptive resolution
2. Natural coupling with gravity
3. Excellent conservation properties
4. Exact advection
5. Easy tracking
6. Easy to model complex geometries
7. Courant timestep free of fluid velocity

Traditional SPH suffer from mixing problems due to suppression of fluid instabilities

Fluid instabilites in reality



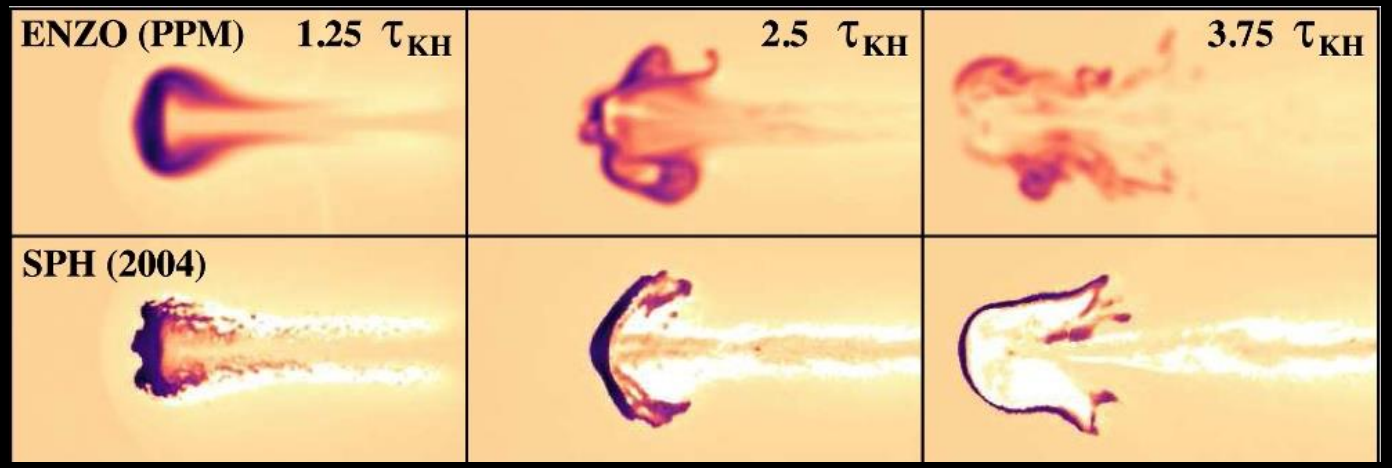
Fluid instabilites in SPH

- Traditional SPH supresses fluid instabilities
  Agertz et al. 2007

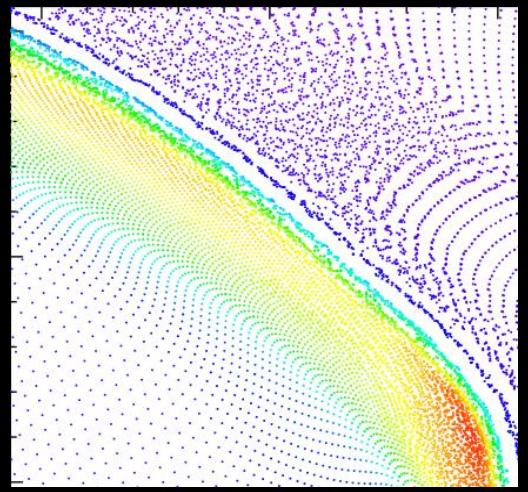
Inconsistent force calculations of particles across contact discontinuities

Grid

Traditional SPH (TSPH)



Surface tension effect



Fluid instabilites in SPH

- Traditional SPH supresses fluid instabilities
  Agertz et al. 2007

Inconsistent force calculations of particles across contact discontinuities

- GDSPH resolves the issue

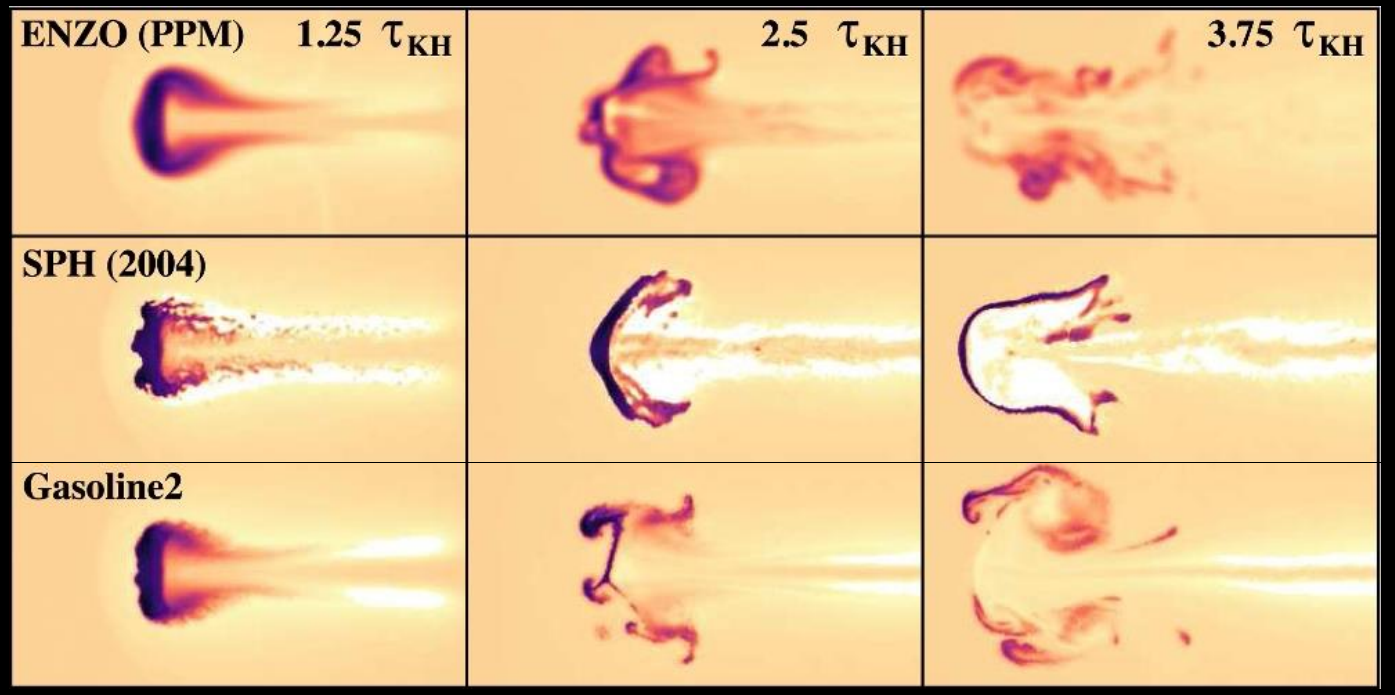
New gradient operator - Geometric average weighting
Wadsley et al. 2017

Turbulent Diffusion terms
Wadsley et al. 2008; Shen et al. 2010

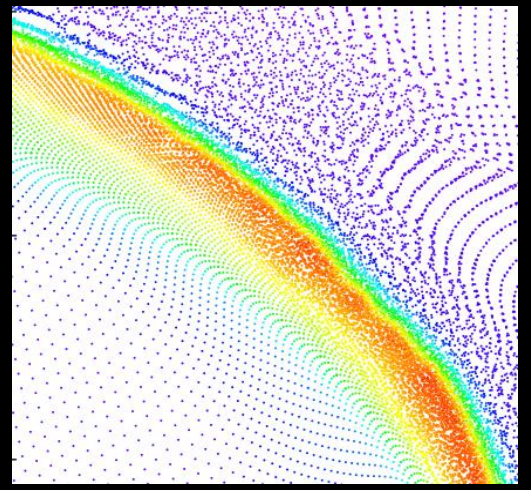
Grid

Traditional SPH (TSPH)

GDSPH



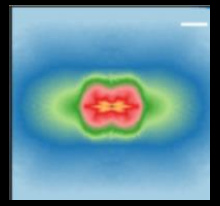
~~Surface tension effect~~

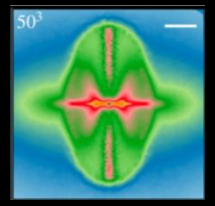


The issues of previous SPH+MHD

- Bad gradient operators

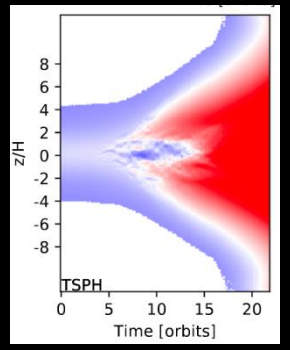
Failure to capture jet formation during magnitized cloud collapse

TSPH 

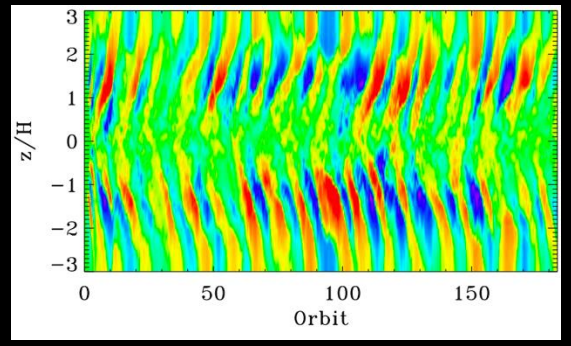
MFM 

Leads to run-away growth in stratified magneto-rotational instability simulations

TSPH



Grid

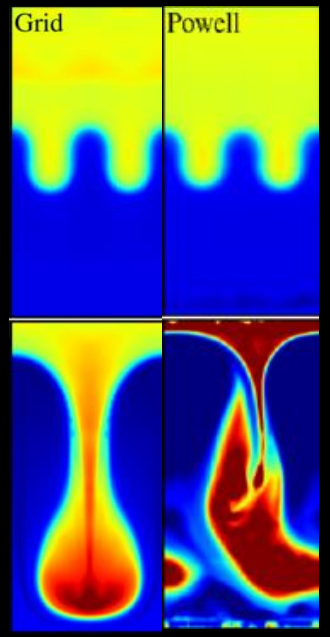


- Divergence error ($\nabla \cdot B \neq 0$)

Lead to flux and force errors

Need to be removed/minimized

Simply advecting divergence error (Powell method) can lead to bad solution in SPH



New SPMHD implementation

- MHD with GDSPH operator

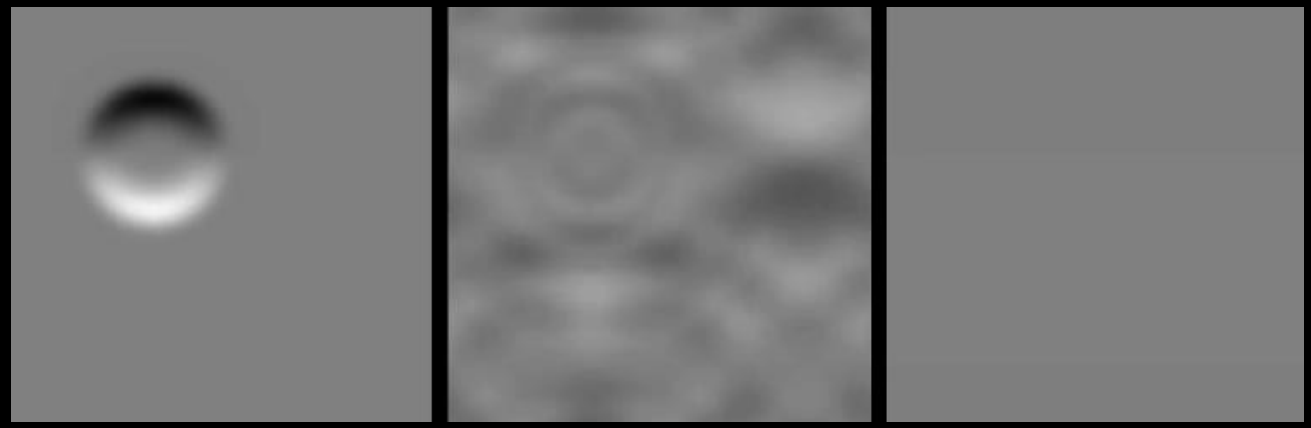
- Constrained hyperbolic divergence cleaning
  Tricco et al. 2016

$$\frac{\partial^2(\nabla \cdot B)}{\partial t^2} - c_h^2 \nabla^2(\nabla \cdot\ ) + \frac{1}{\tau}\frac{\partial(\nabla \cdot B)}{\partial t} = 0$$

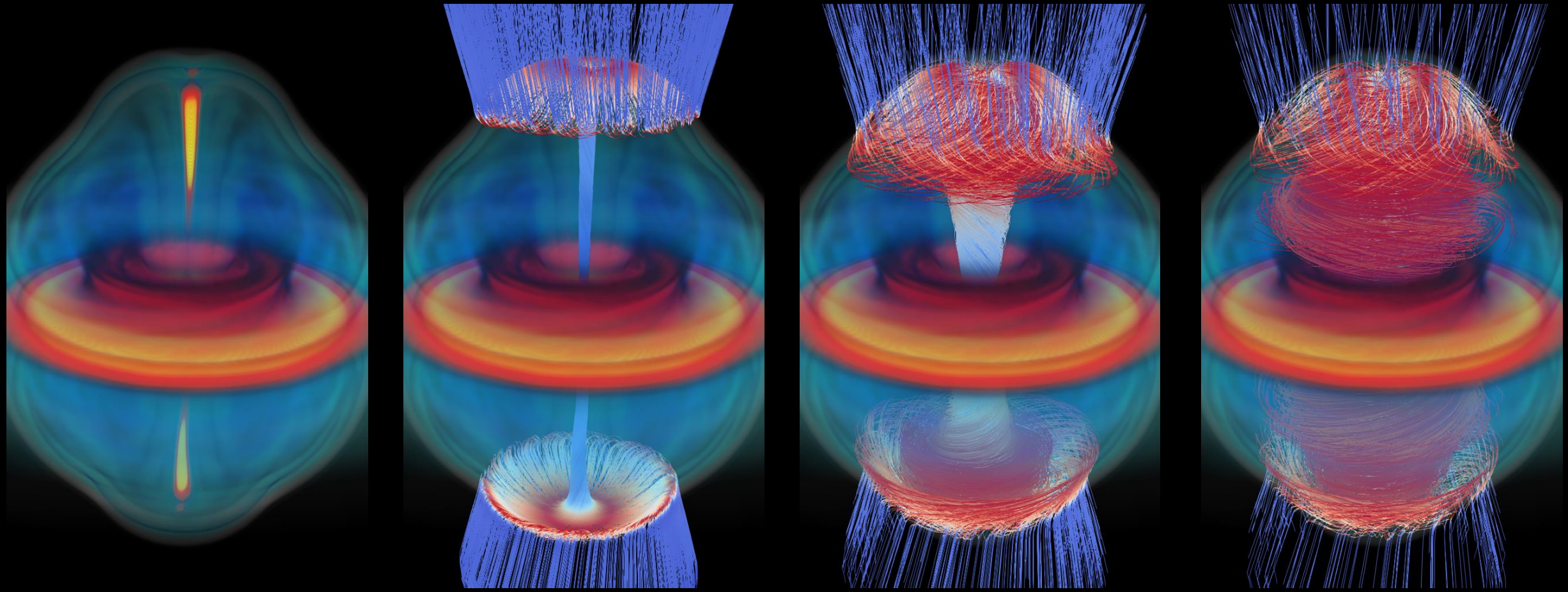
Divergence error evolved as a damped wave

Energy conserving formulation of divergence cleaning (stable at discontinuities)

Static
$\nabla \cdot B$ test

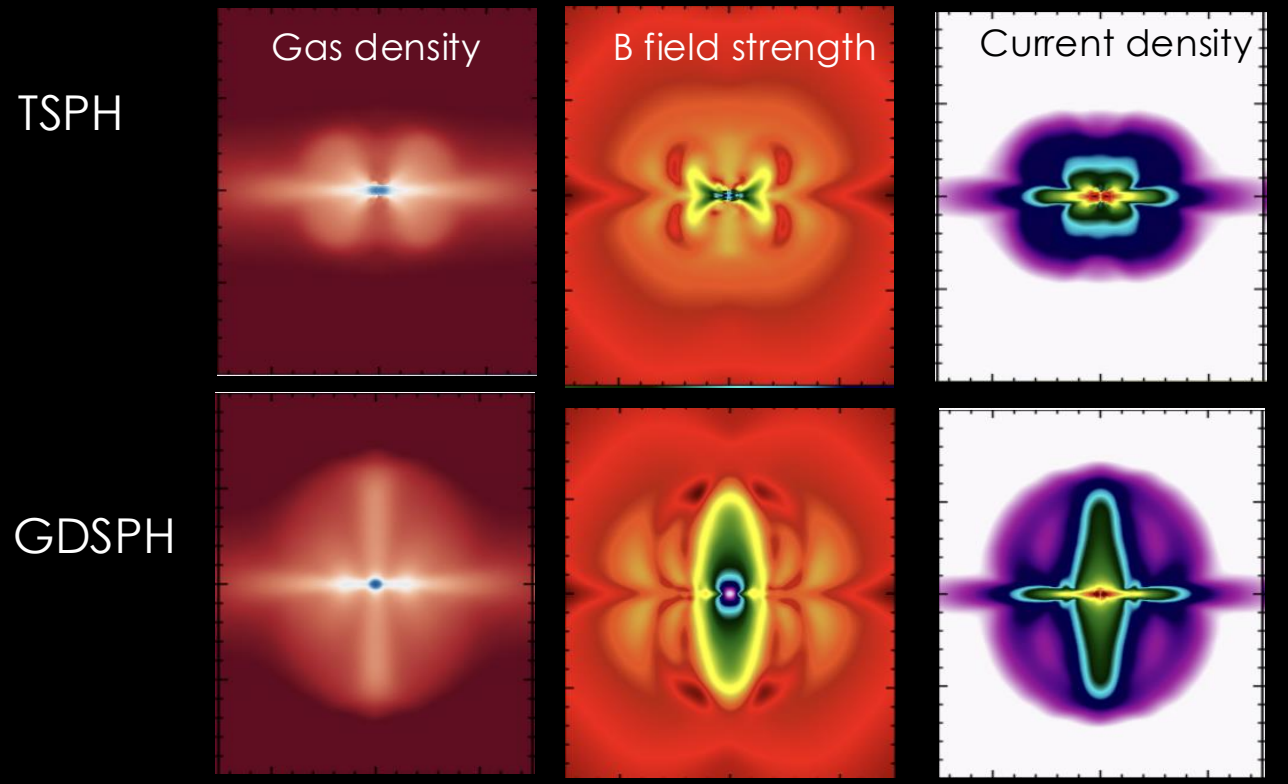


Magnetized core collapse



Magnetized core collapse

- Jet formation captured at much lower resolution and for weaker magnetic fields
- Similar results to MFM/MFV in Gizmo in cloud collapsing/jet launching (Hopkins & Ravis 2016)



Wissing & Shen (2020), A&A 638, A140

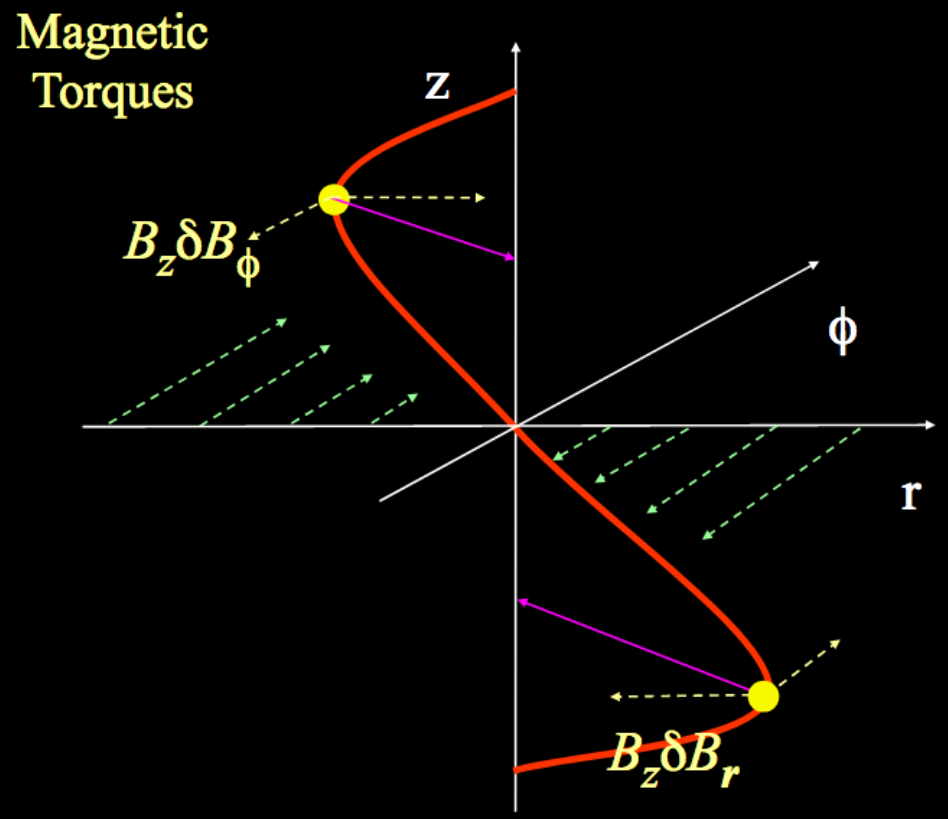
Magnetorotational instability (MRI)

- Local linear instability that occurs in disks satisfying:

$$\frac{d\Omega^2}{d\ln R} < 0$$

$$k^2 \frac{B^2}{\rho} < -\frac{d\Omega^2}{d\ln R}$$

- Requires sufficiently ionized gas
- Leads to exponential growth of the field



Credit: Martin Pessah

Magnetorotational instability (MRI)

- Local linear instability that occurs in disks satisfying:

$$\frac{d\Omega^2}{d\ln R} < 0$$

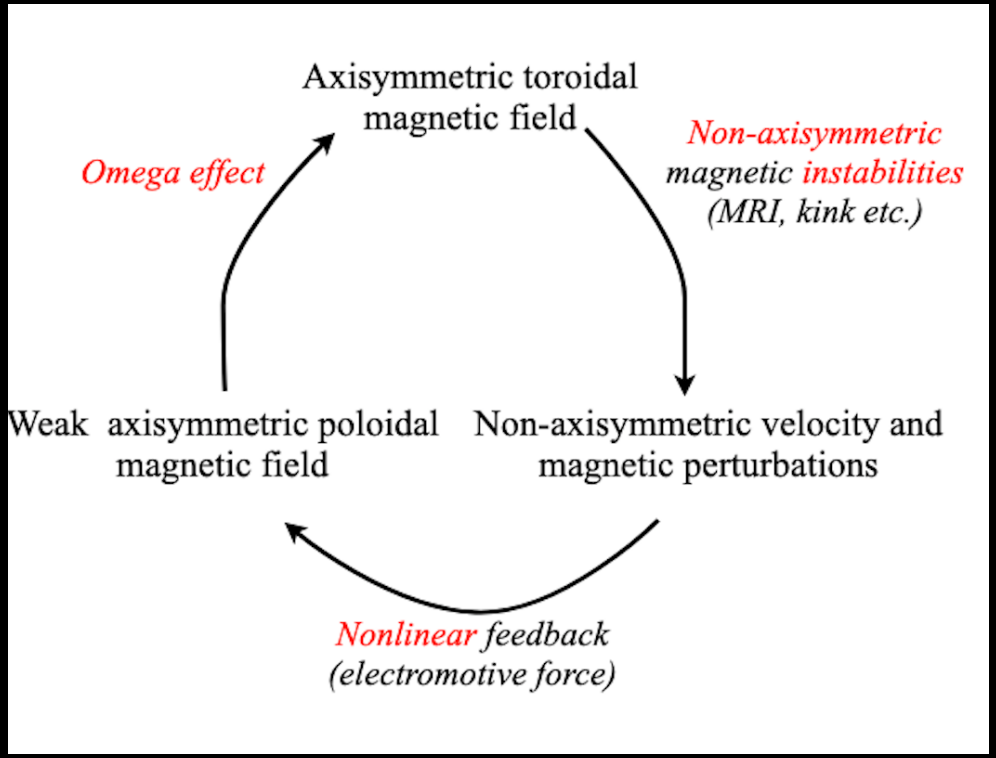
$$k^2 \frac{B^2}{\rho} < -\frac{d\Omega^2}{d\ln R}$$

- Requires sufficiently ionized gas
- Leads to exponential growth of the field

- The instability eventually breaks down into MHD turbulence

Requires self-sustaining process to remain/saturate

Driven by non-linear dynamics and dynamo effects



Riols et al. 2013

Magnetorotational instability (MRI)

- Local linear instability that occurs in disks satisfying:

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$$k^2 \frac{B^2}{\rho} < -\frac{d\Omega^2}{d\ln R}$$

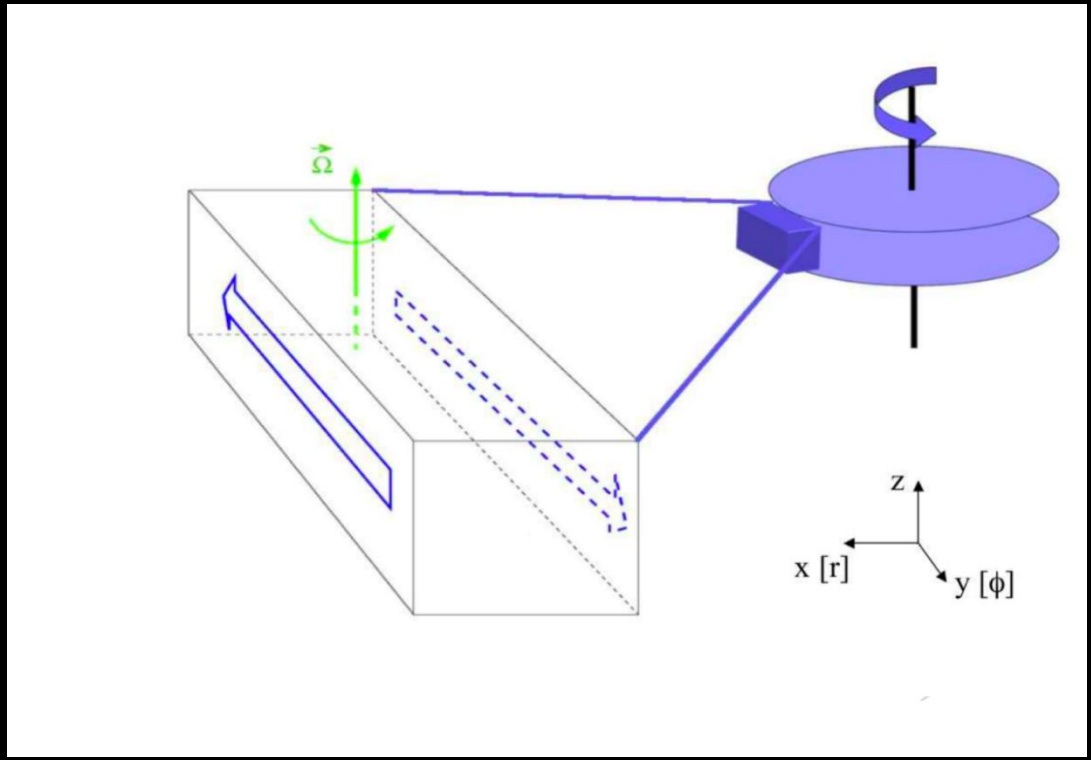
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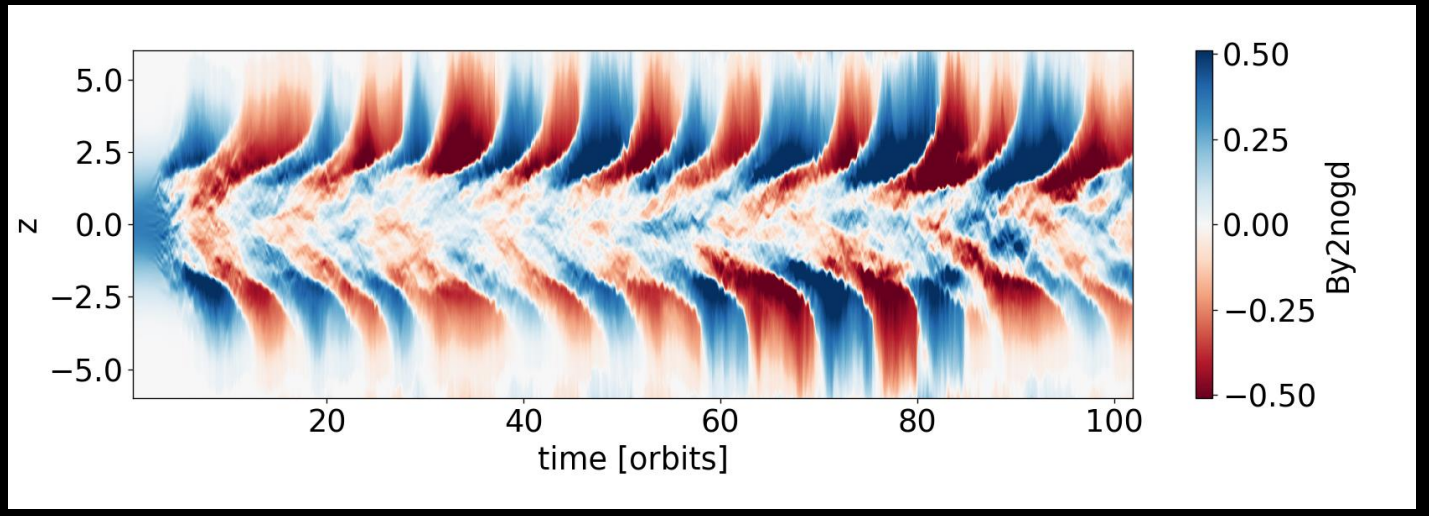
- Ensued turbulence produce stress(α_{stress}) in disks and act as a driver of angular momentum transport
- Local simulations studied with shearing boxes



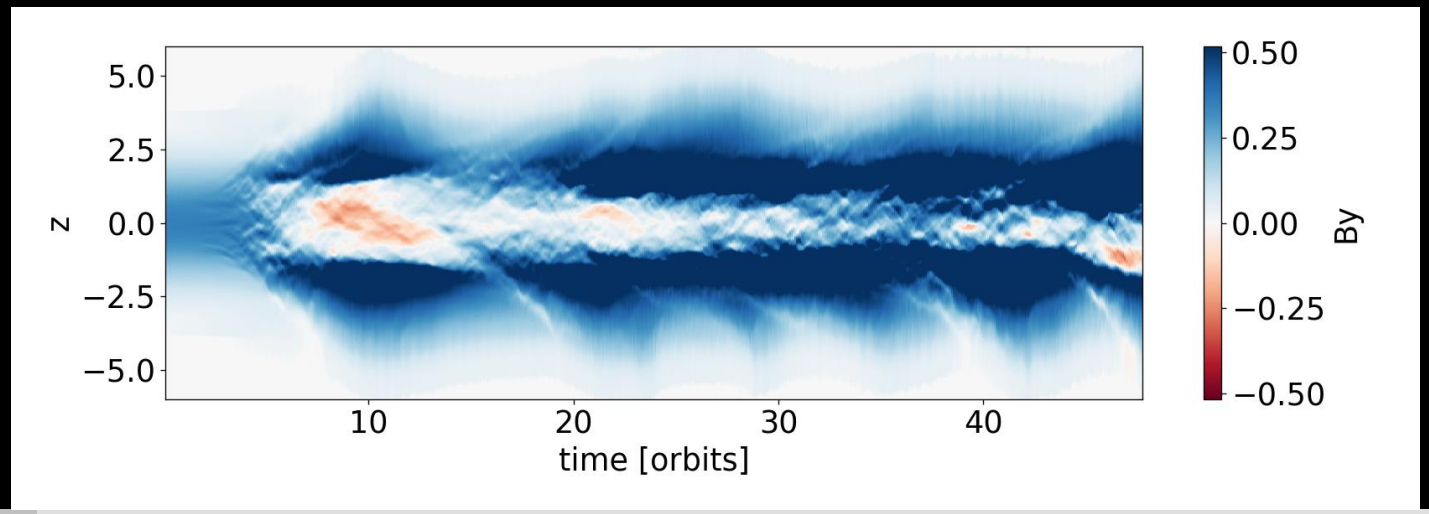
Credit: Lin Ming

MRI in stratified medium

GDSPH

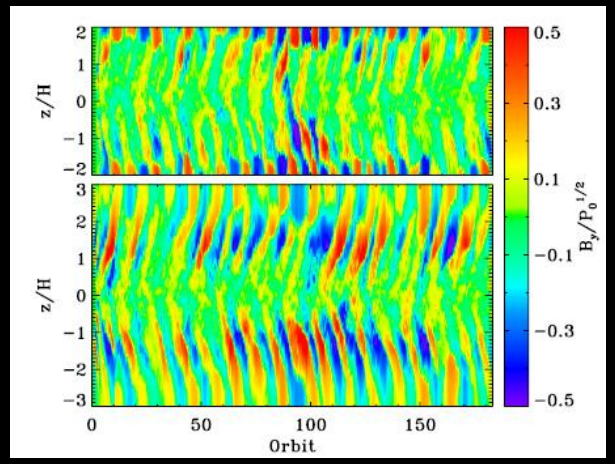


TSPH

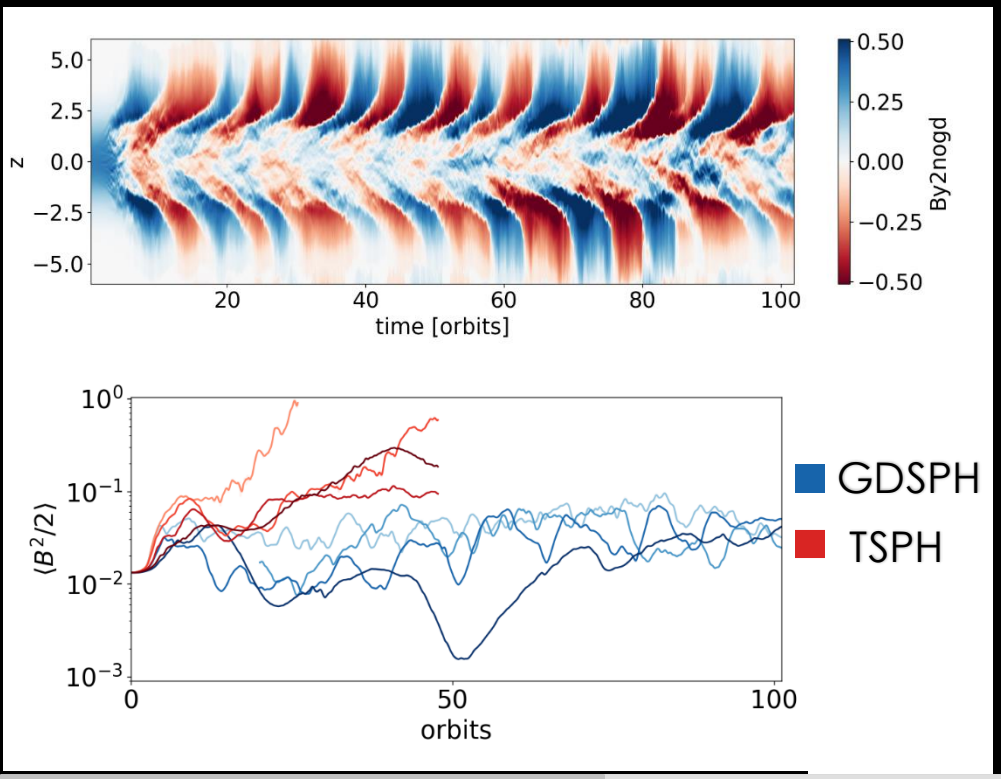


Comparison to other codes

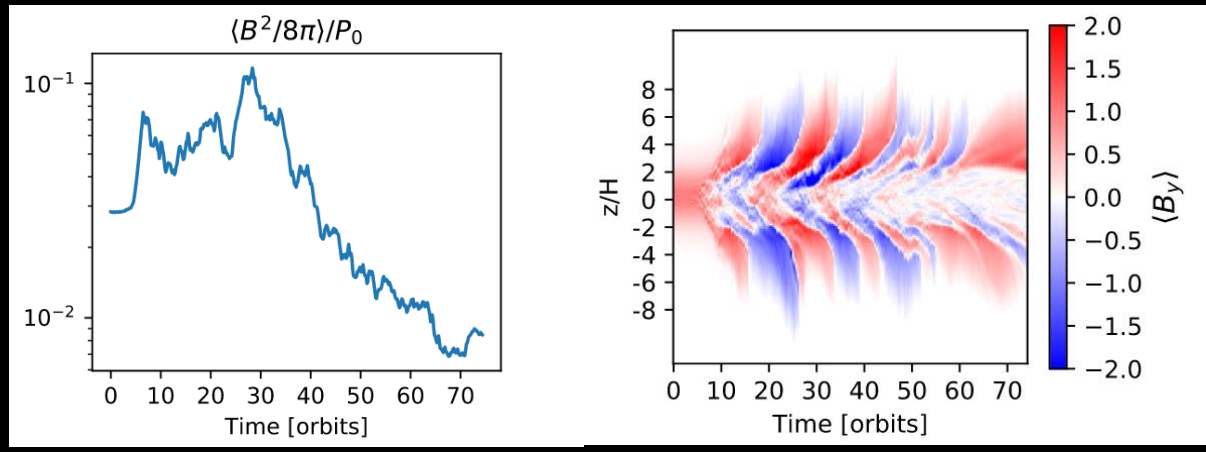
ATHENA



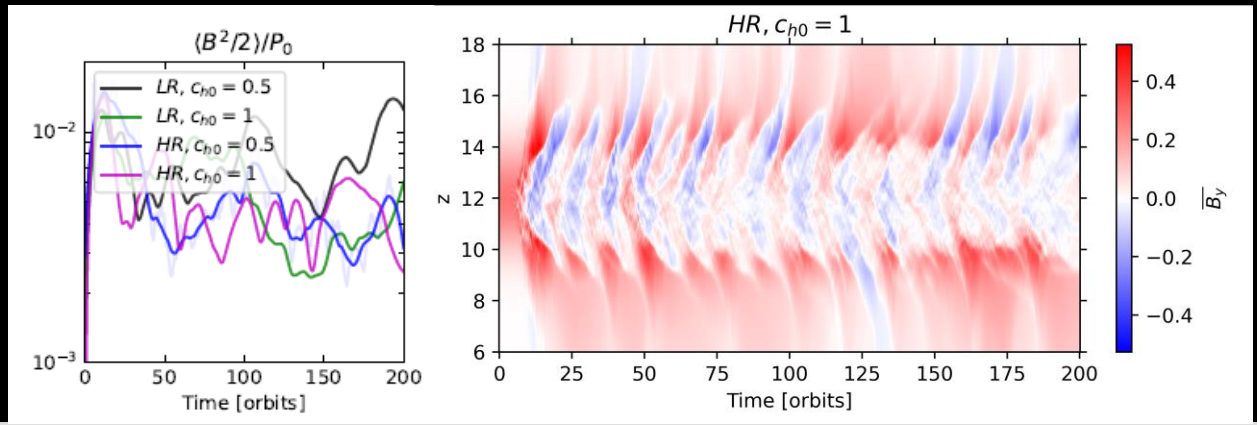
GDSPH



MFM



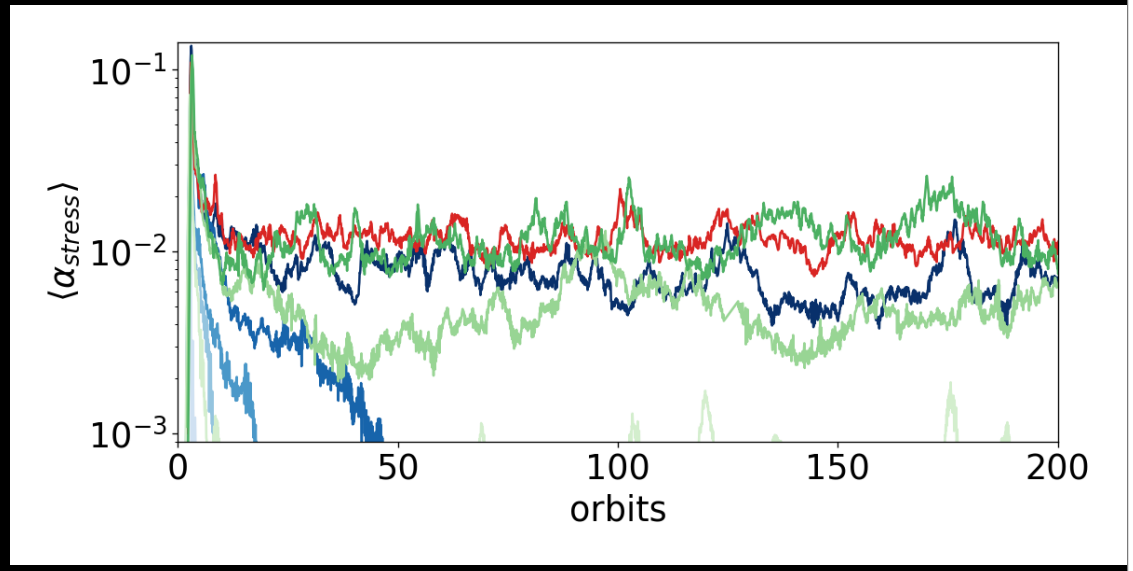
AREPO



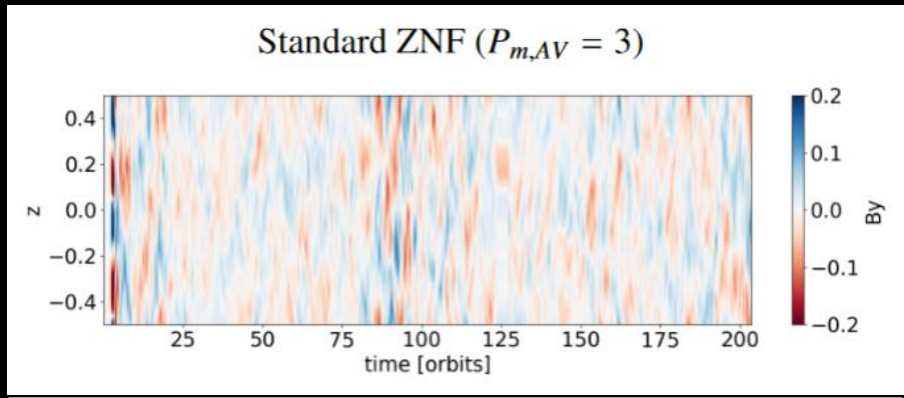
MRI in unstratified medium

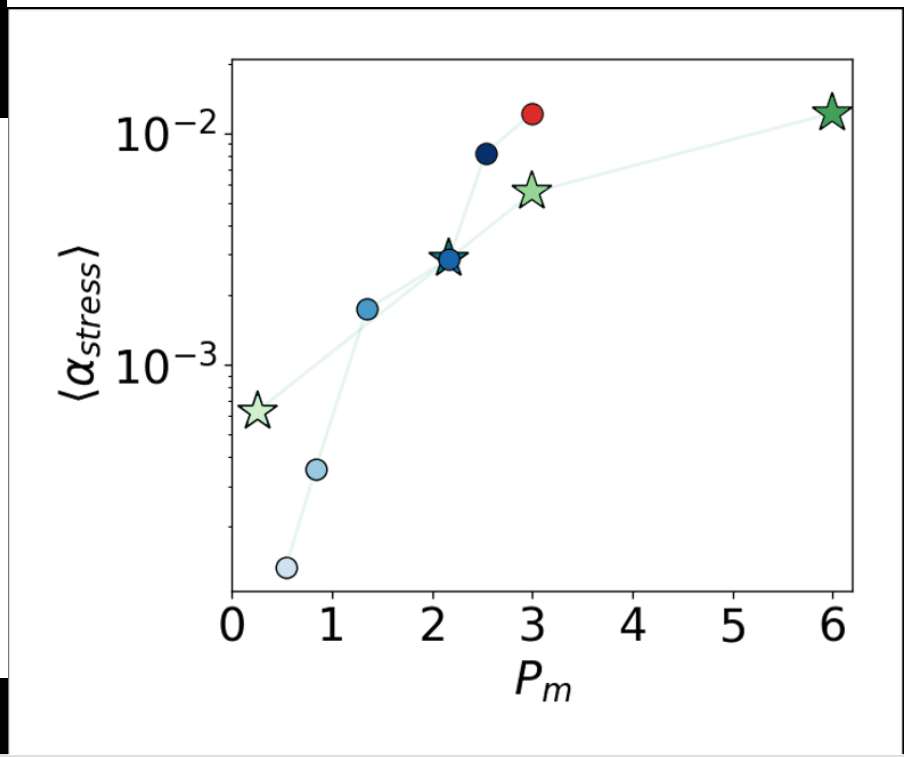
- Saturates if above critical Prandtl number $P_m \sim 2.5$, similar to grid codes
- Increase in stress(α_{stress}) with Prandtl number
- Stress independent of resolution at given Prandtl number

$$P_m = \frac{viscosity}{resistivity} = \nu/\eta$$



Wissing et al. 2021





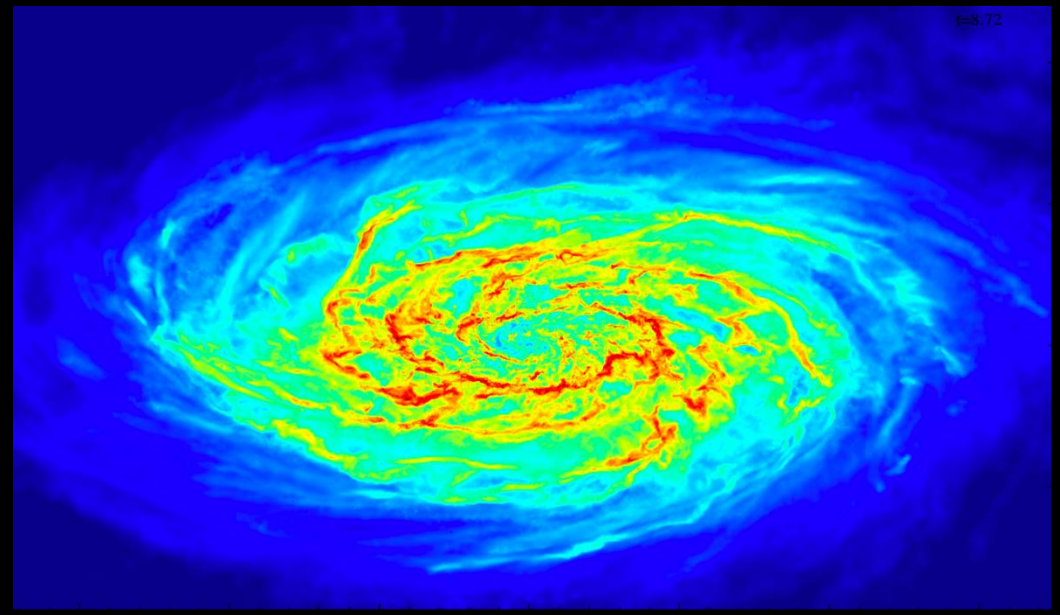
Galaxy Simulations

- 53 Isolated disk from AGORA comparison project, MW-like disks (Kim+2015);
- UVB (z = 0), radiative heating/cooling; star formation, SN feedback;
- Parameter space:
  - Resolution (low, medium, high)
  - Jeans floor (minimum collapsing length);
  - Feedback models:
    - delayed cooling (Stinson+2006) vs. superbubble (Keller+2014);
    - Different injection length;
    - Feedback strength
  - Diffusion parameters;
  - Initial B field configuration;

Wissing & Shen (2022)

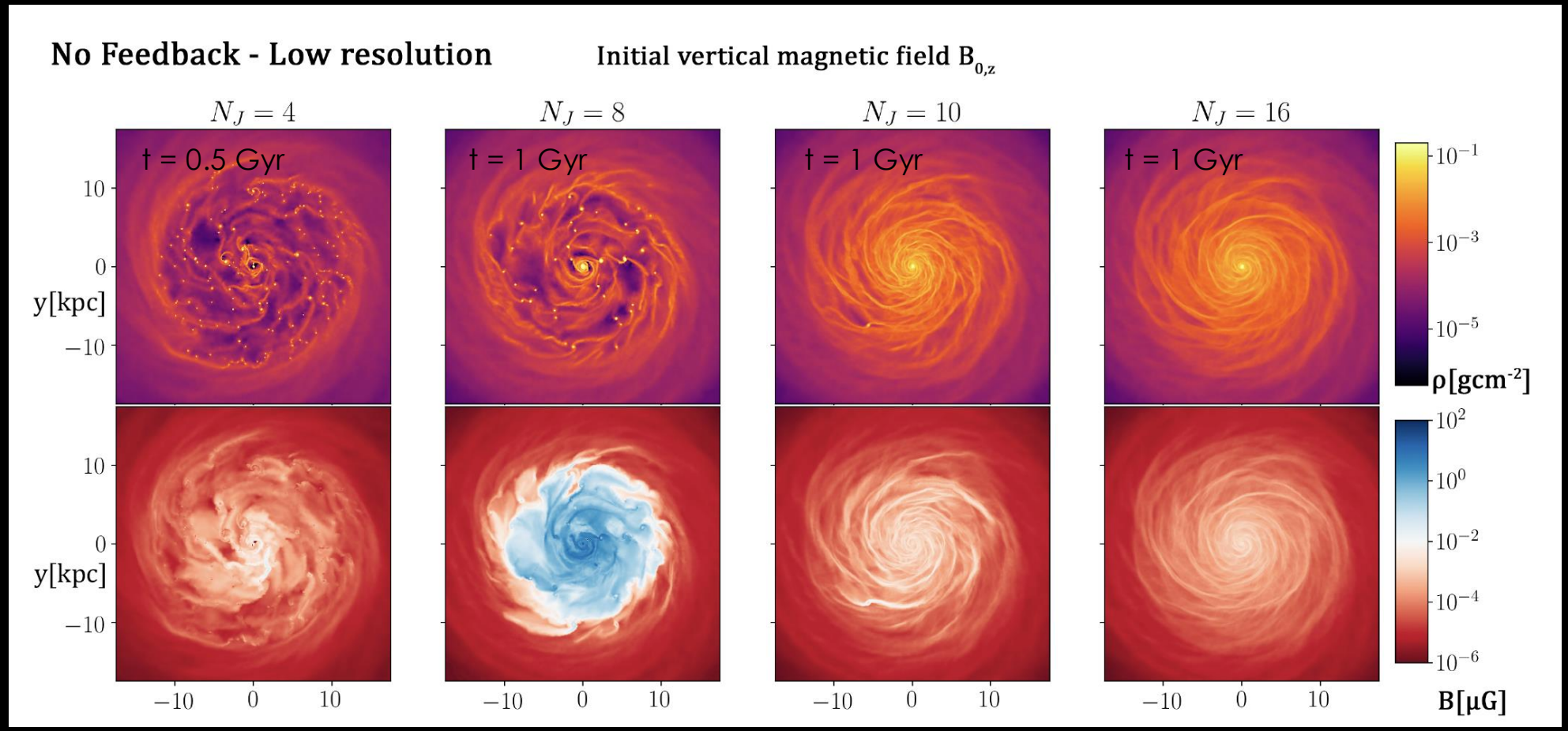
	N_{tot}	$m_{gas}[M_\odot]$	$m_{star}[M_\odot]$	$m_{dark}[M_\odot]$	$\epsilon_{gas}[kpc]$
Low	$2.5 \cdot 10^6$	$1.0 \cdot 10^4$	$4.3 \cdot 10^4$	$1.6 \cdot 10^6$	0.02
Medium	$2.0 \cdot 10^7$	$1.3 \cdot 10^3$	$5.4 \cdot 10^3$	$2.0 \cdot 10^5$	0.01
High	$1.6 \cdot 10^8$	$1.7 \cdot 10^2$	$6.7 \cdot 10^2$	$2.5 \cdot 10^4$	0.005

Typical “high-resolution” for cosmological zooms



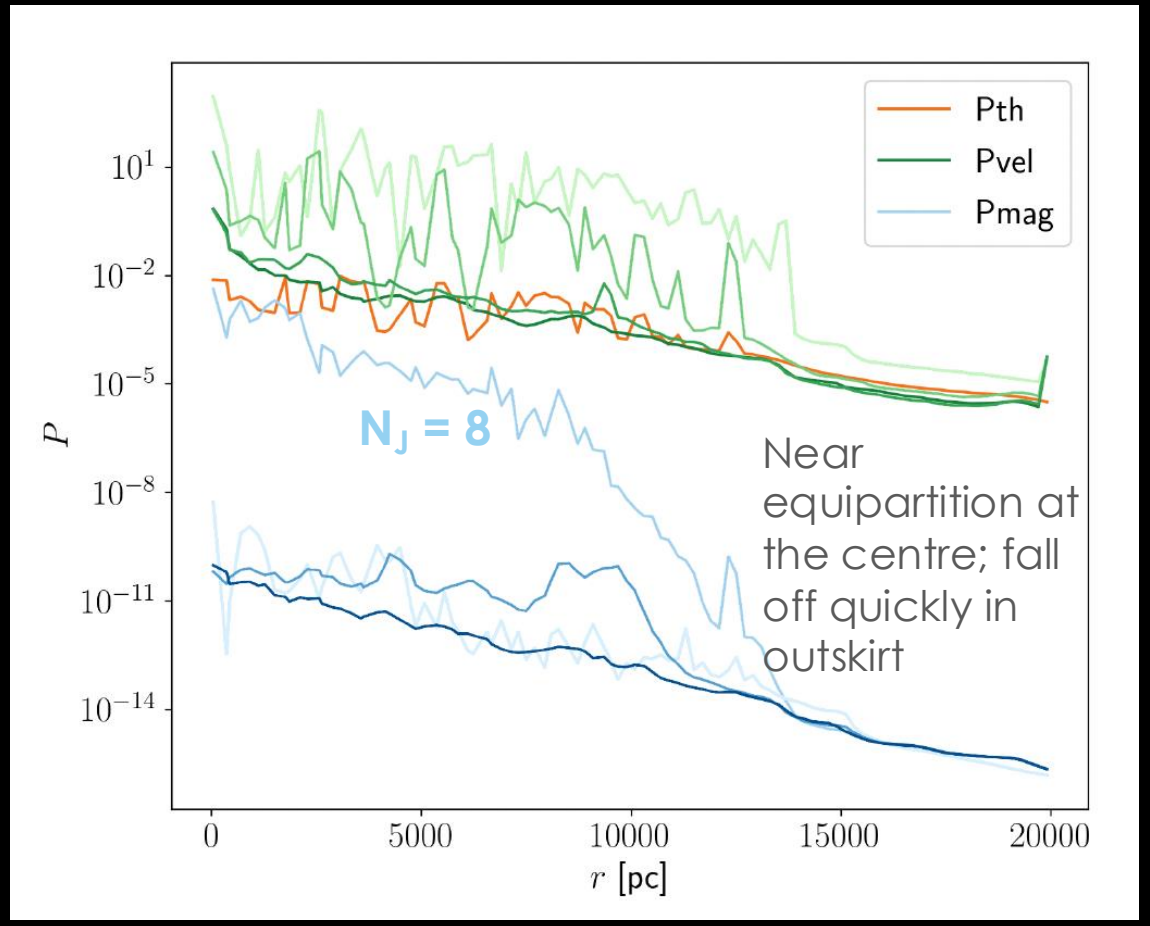
No feedback – GI, collapse and shear

Gas compression in very unstable discs amplify B field rapidly

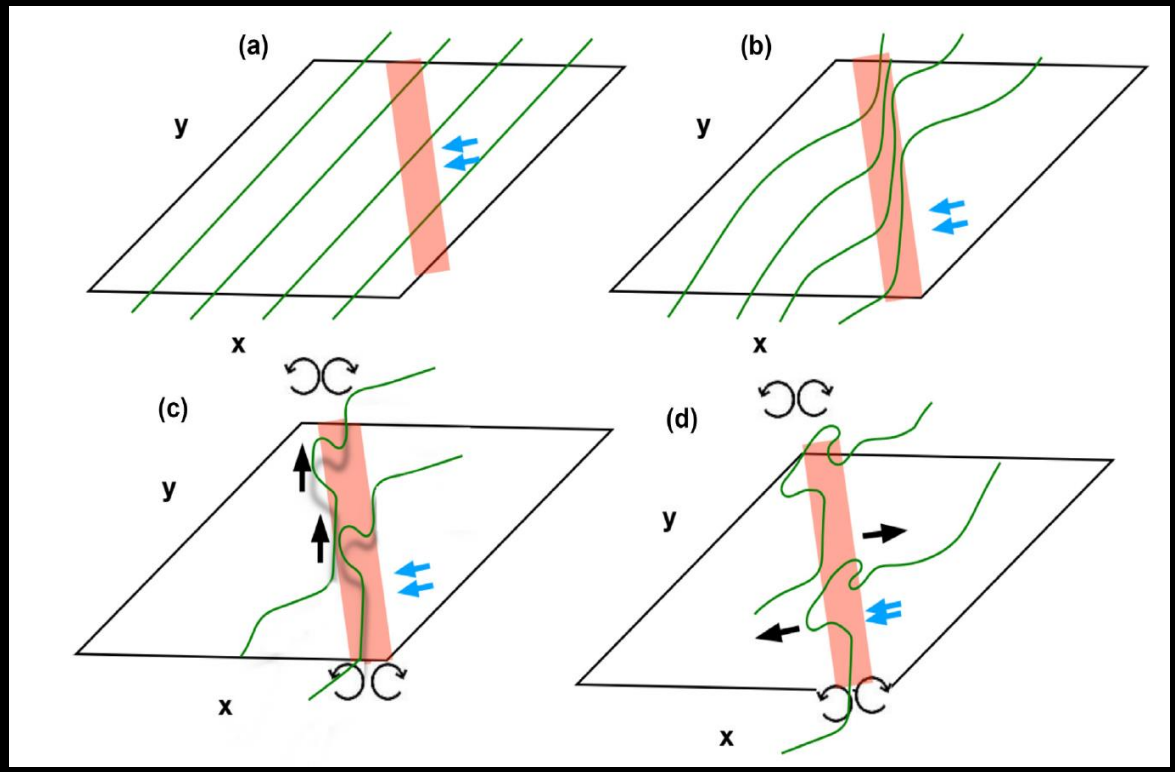


Larger Jeans support, weaker gravitational instability;

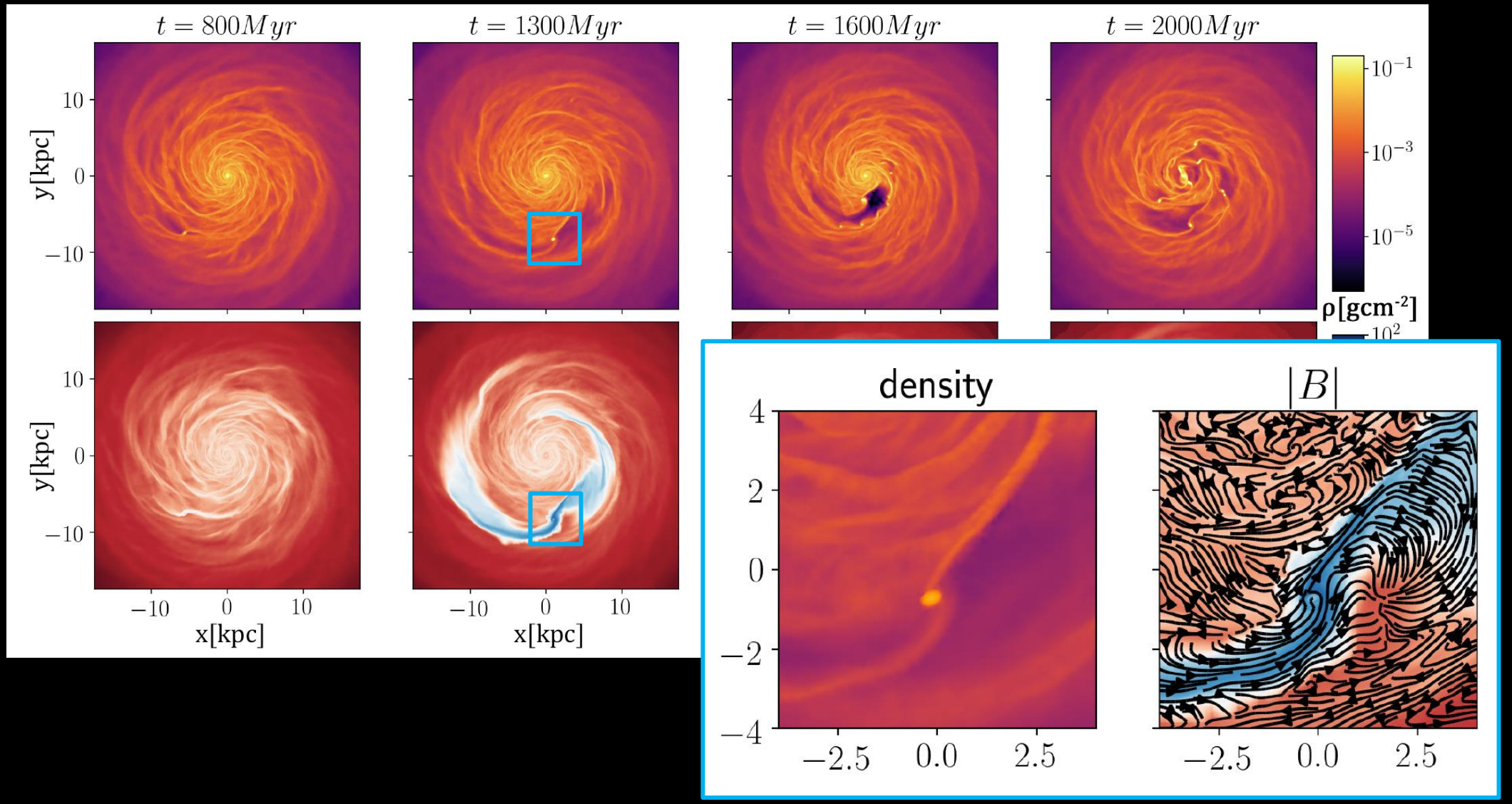
No feedback: GI dynamo?



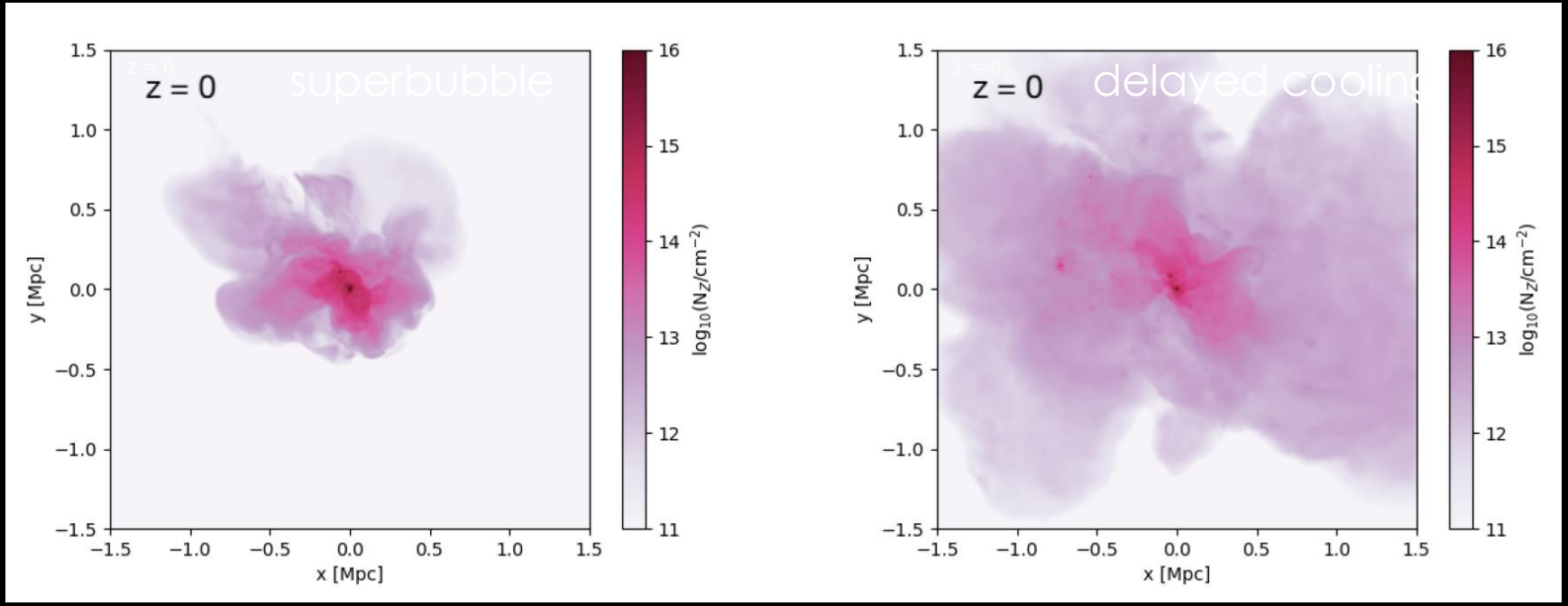
- Strong GI leads to amplification, similar to protoplanetary discs (e.g., Rios & Latter 2019)
- Depletion of gas ($N_J = 4$) leads to decaying of B field;



No feedback: GI dynamo



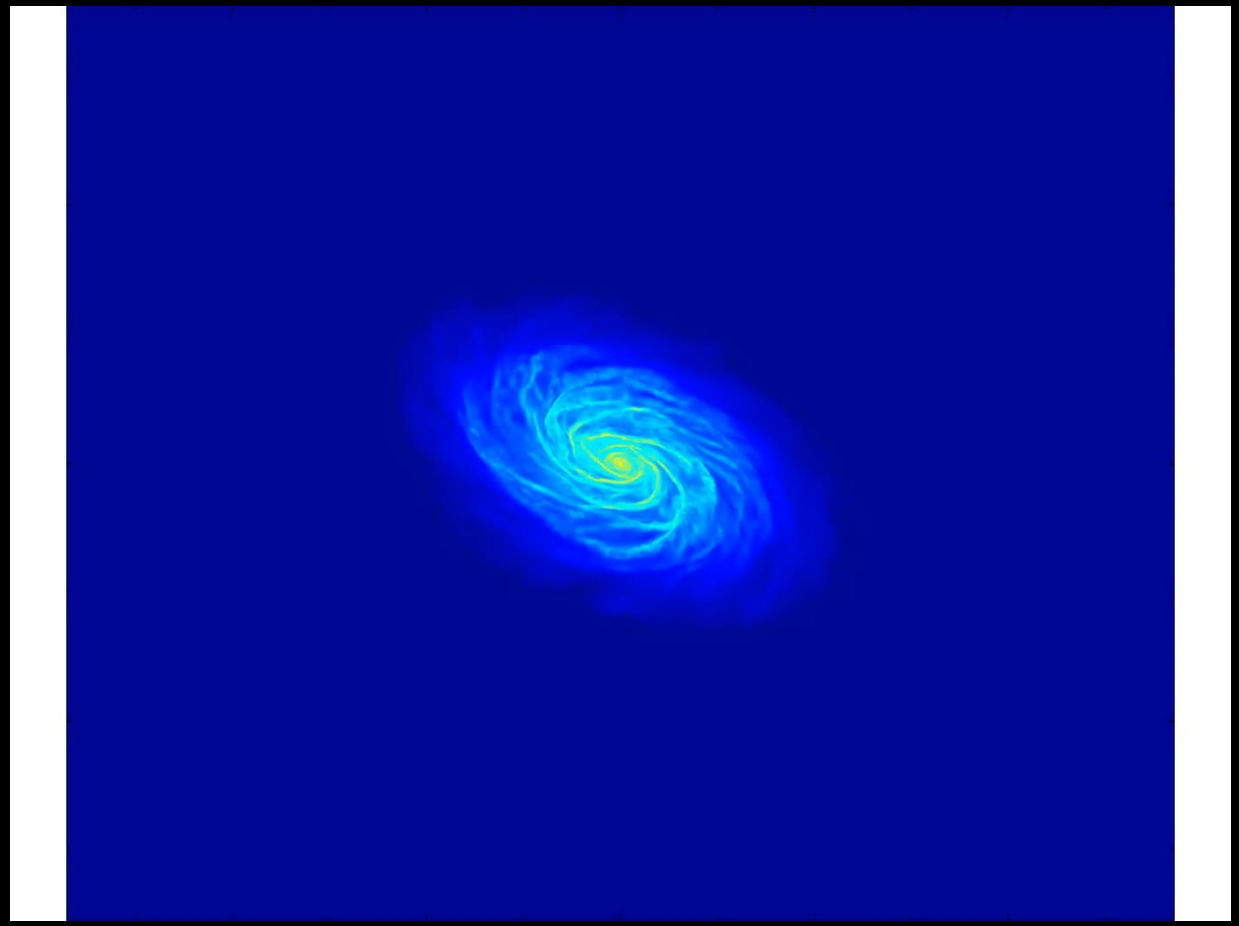
Adding feedback (subgrid)



Mina, Shen+ (2021)

- Delayed cooling approach (e.g. Stinson+2006; Agertz+2013): radiative cooling temporary delayed ~ 10 Myr or so >> Sedov-Taylor expansion phase;
- Superbubble model (Keller+2015): SN clustering, subgrid thermal conduction governing ISM mass loading

B-field amplified by feedback

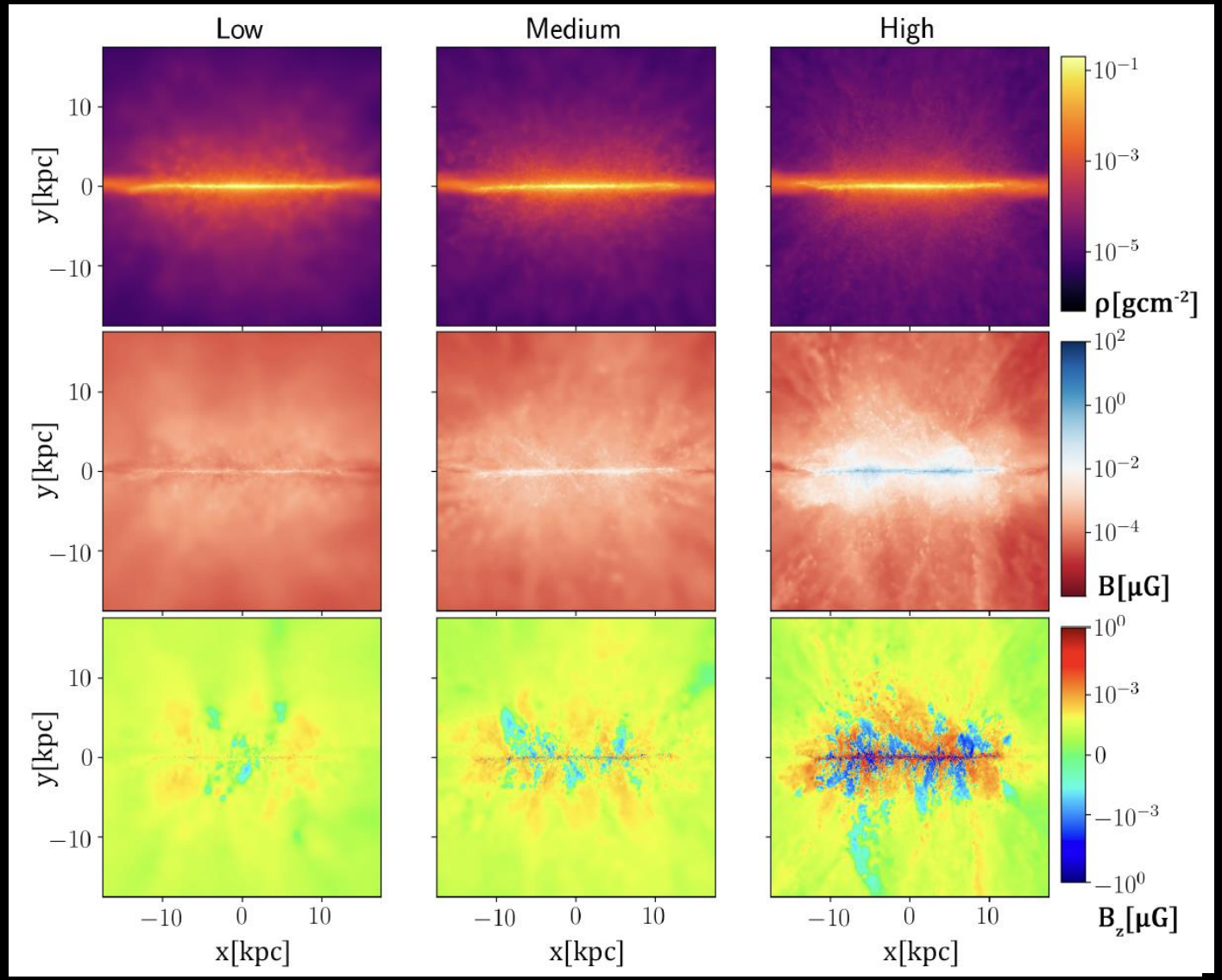


- Superbubble model, with N_{FB} = 1, E_{SN} = 10^{51} ergs, medium resolution (reaches saturation ~ 1.2 Gyr)
- Feedback
  - Advect/diffuse magnetic field, drag into the CGM (destructive esp. central region)
  - Outflow/fountain increase vetical motion -> increase $\alpha\omega$ dynamo
- Magnetization of the CGM

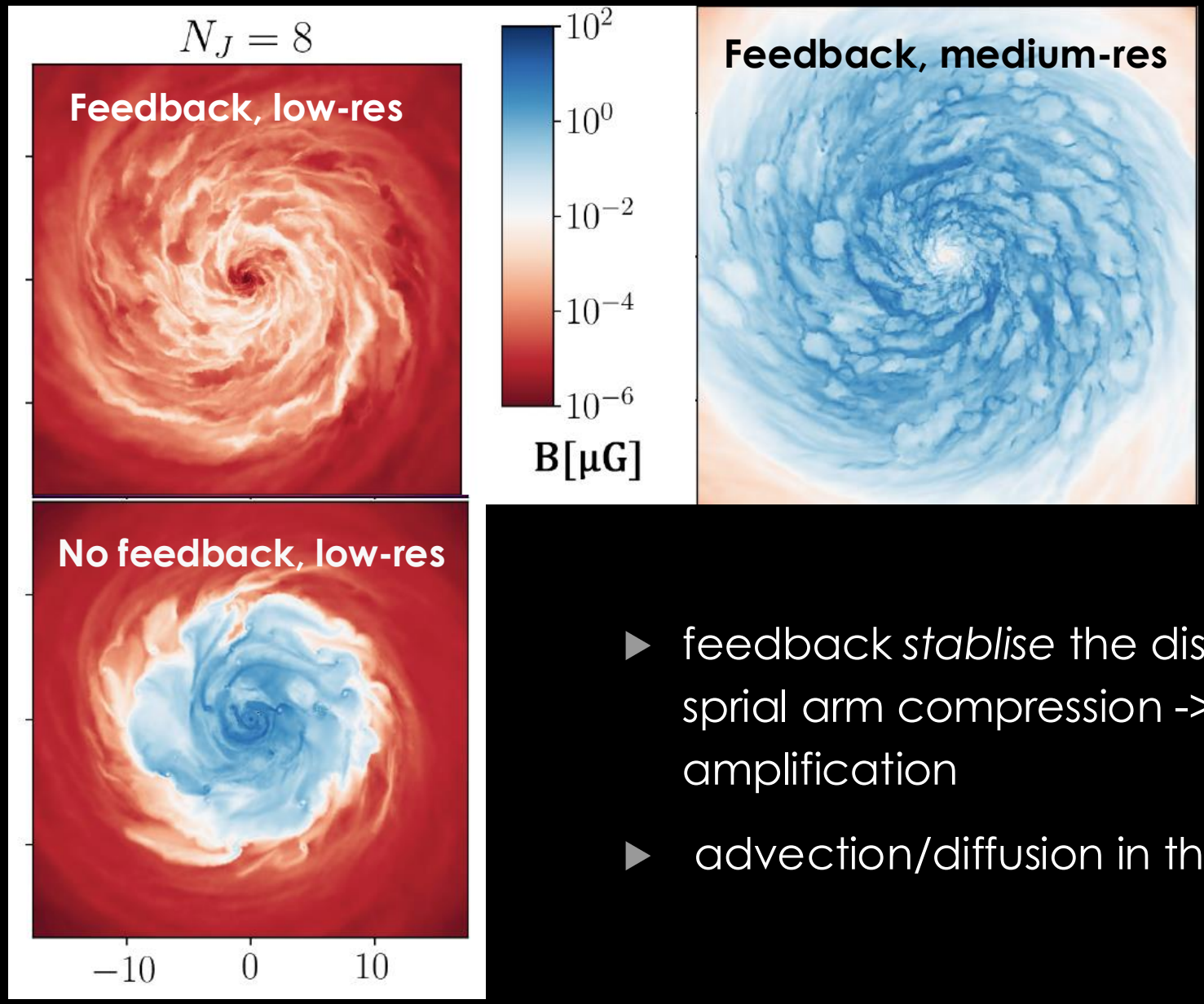
Resolution dependence

Early time t = 250 Myr

- Superbubble model, with N_{FB} = 1, E_{SN} = 10^{51} ergs
- Higher resolution -> faster amplification (not surprising…)
- Polarities and large-scale coherent B-field in all cases-> mean field dynamo (likely still $\alpha\omega$ dynamo)
- Complexities in the CGM (near disc plane) shown in higher resolution



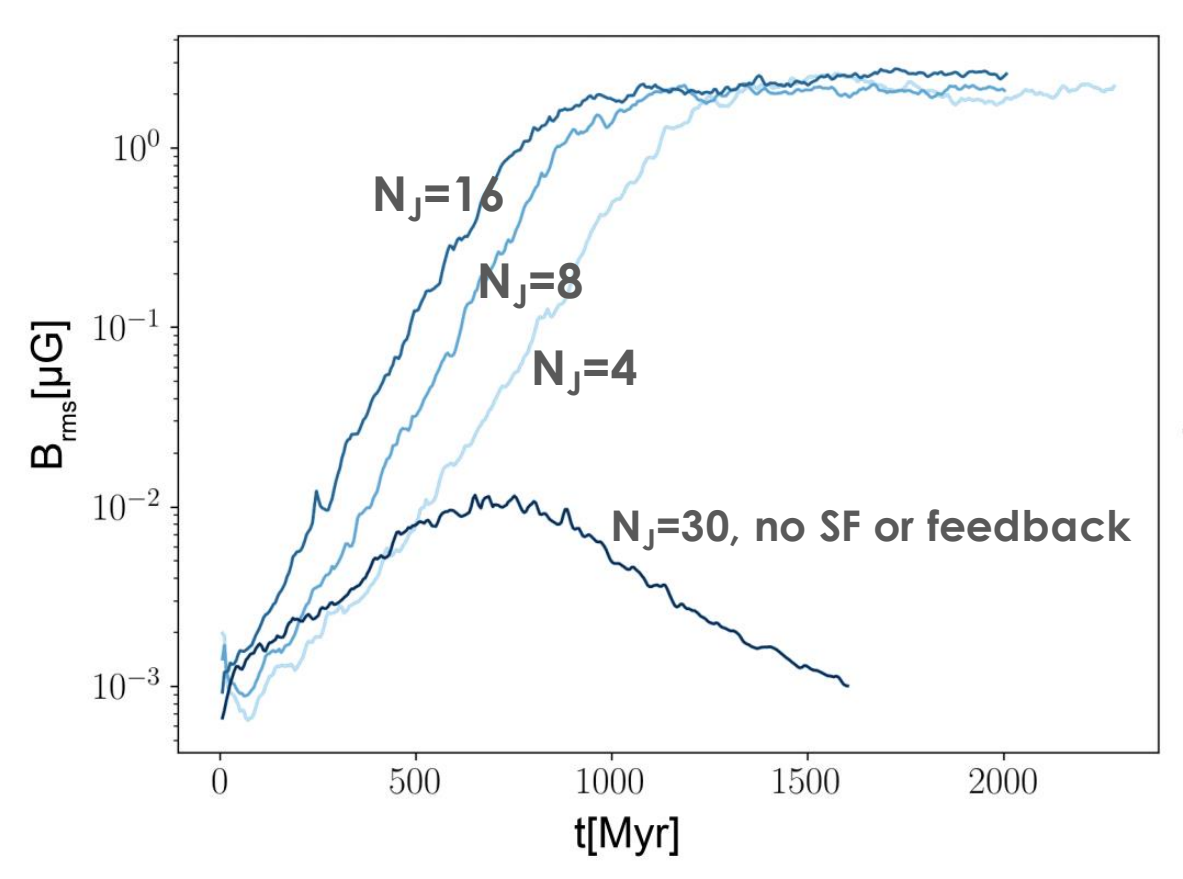
The negative & positive impact of feedback



- But once one resolves the vertical structure better -> boosted amplification with feedback, esp. outer region

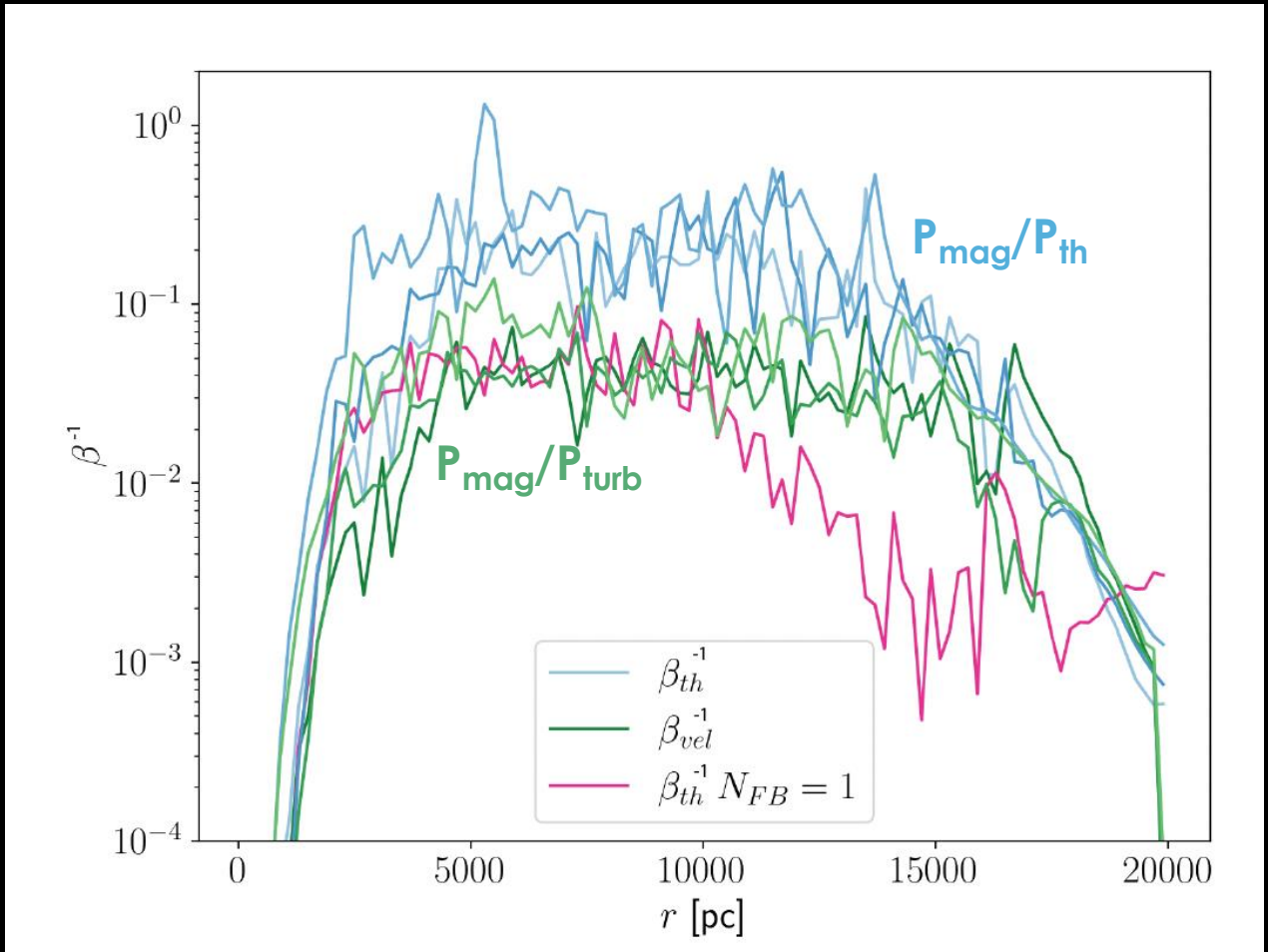
- feedback *stablise* the disc, reduce GI and sprial arm compression -> less amplification
- advection/diffusion in the central region;

Saturation levels and dependency on Jeans support



- Medium resolution with feedback runs saturate to a few μG, regardless the minimum collapsing scale;
- Larger Jeans floor -> faster amplification.

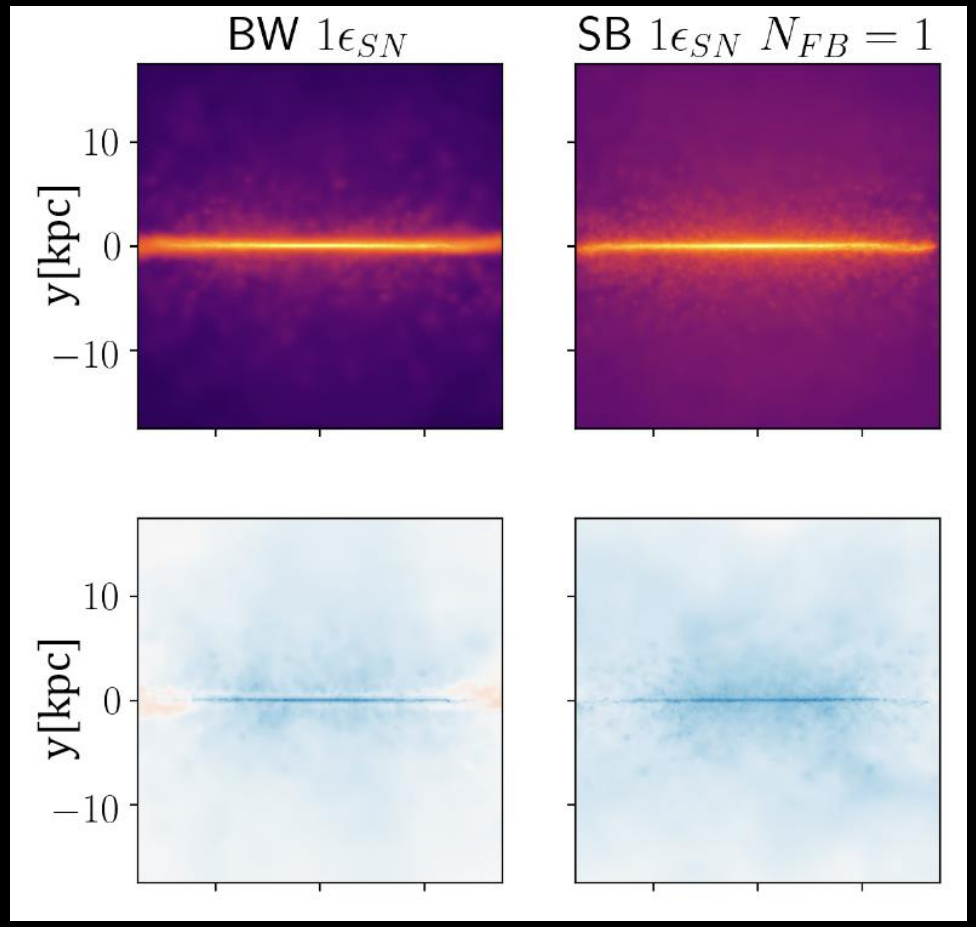
Saturation level



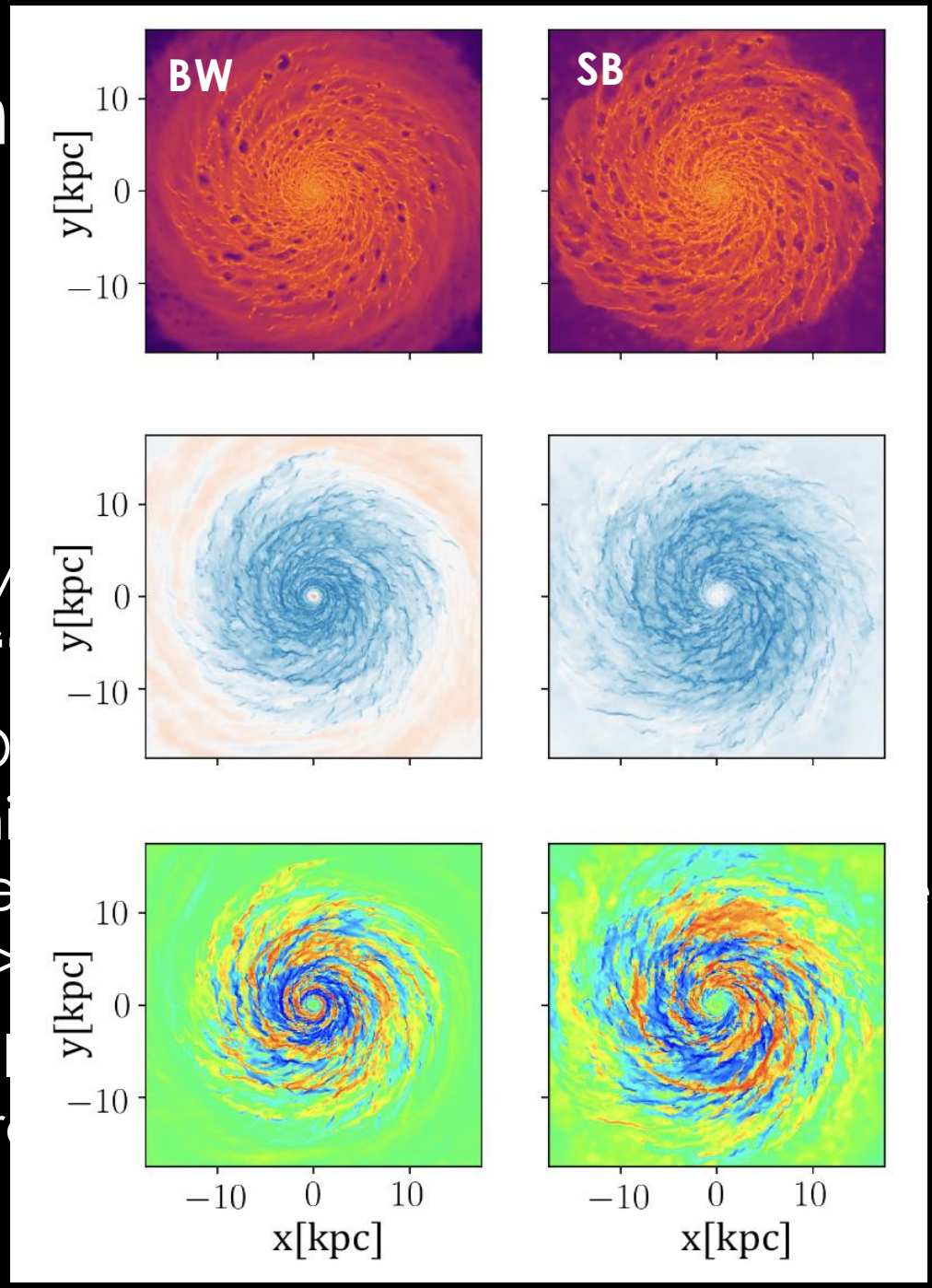
$\beta_x^{-1} \equiv P_{mag}/P_x$

- Magnetic pressure over thermal or turbulent pressure remain flat for 4-15 kpc
- Average saturate around 10-30% thermal pressure and ~ 5% of kinetic pressure;
- B field strength similar to that of the MW

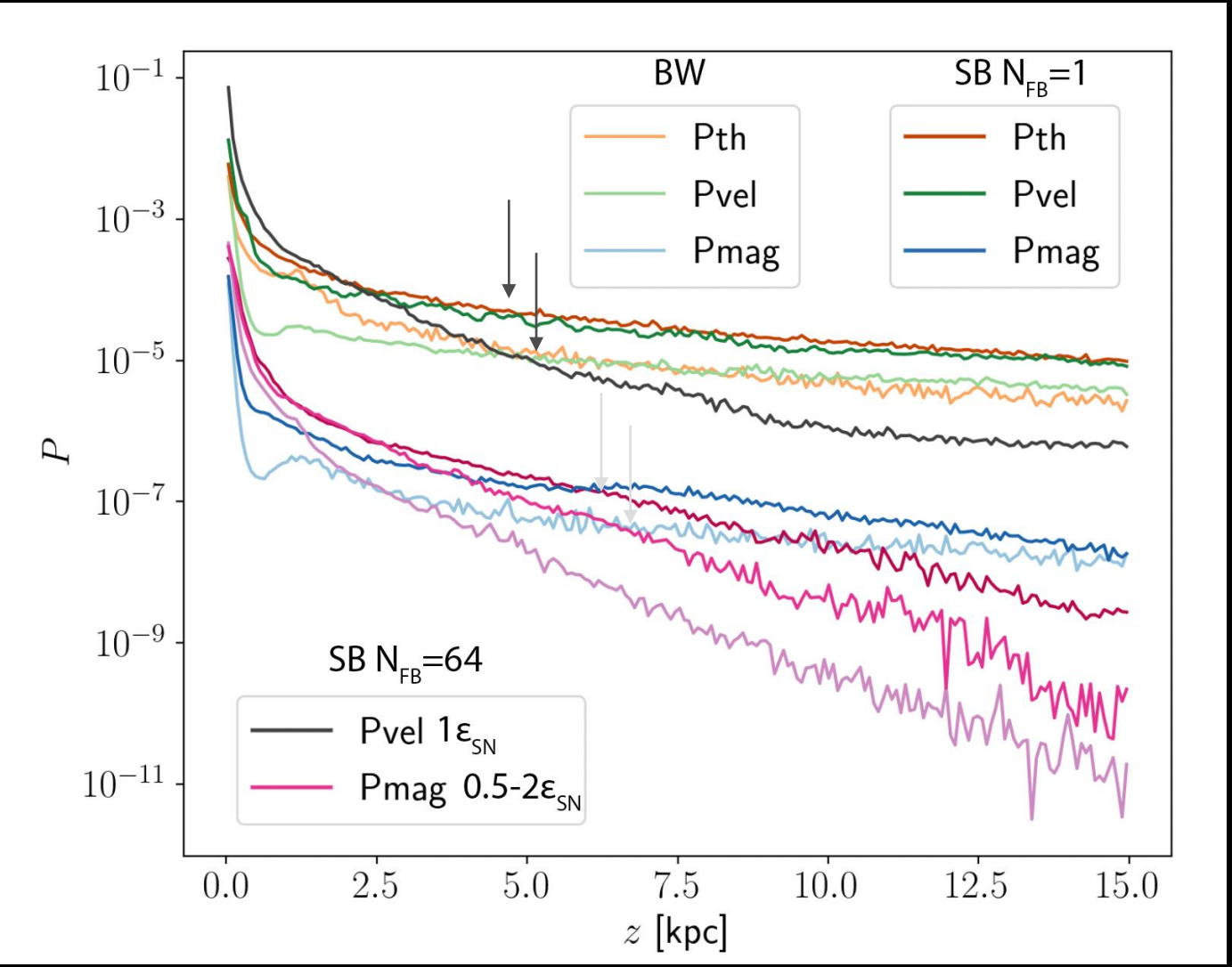
Variation on feedback m



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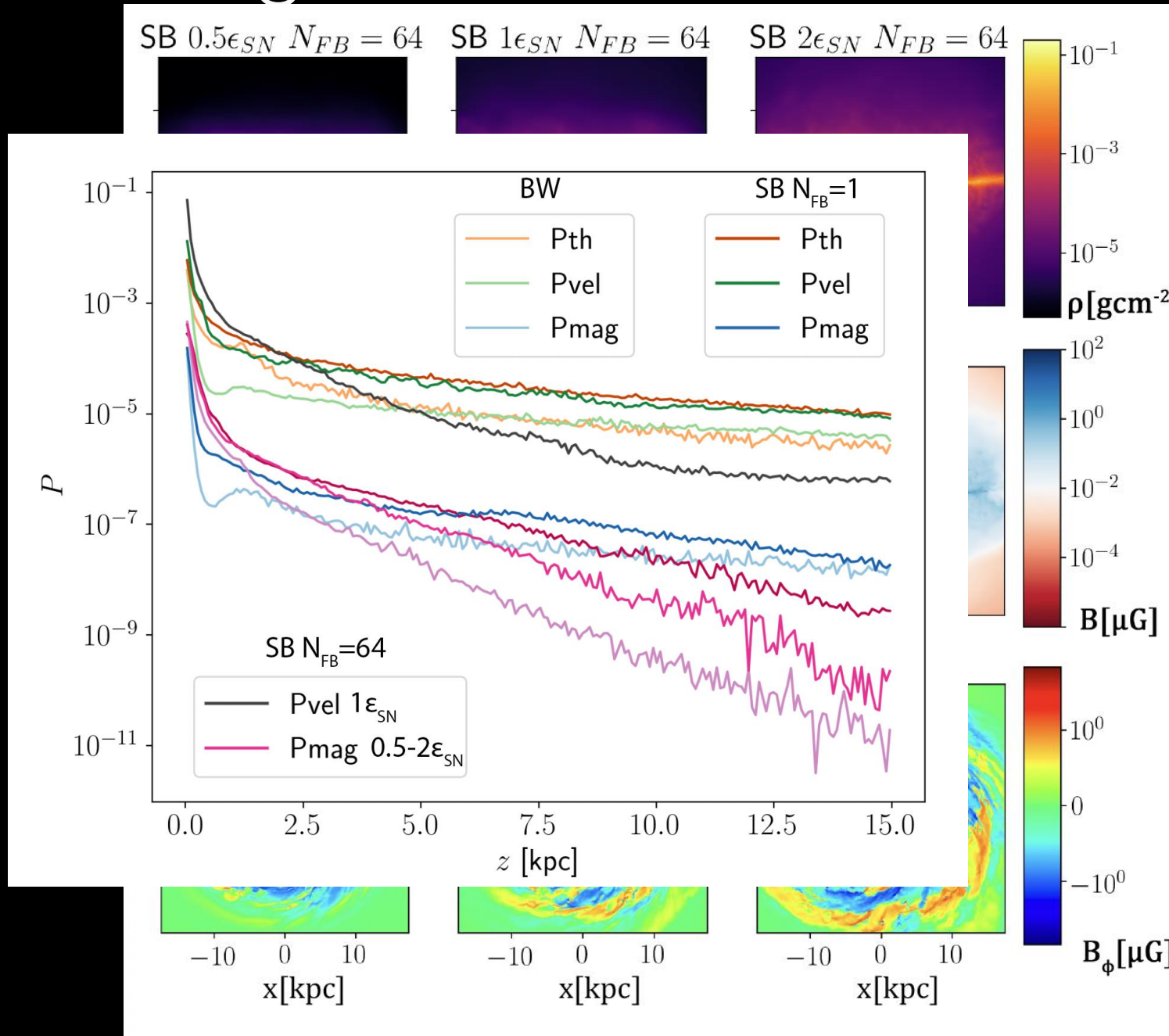
Variation on feedback models



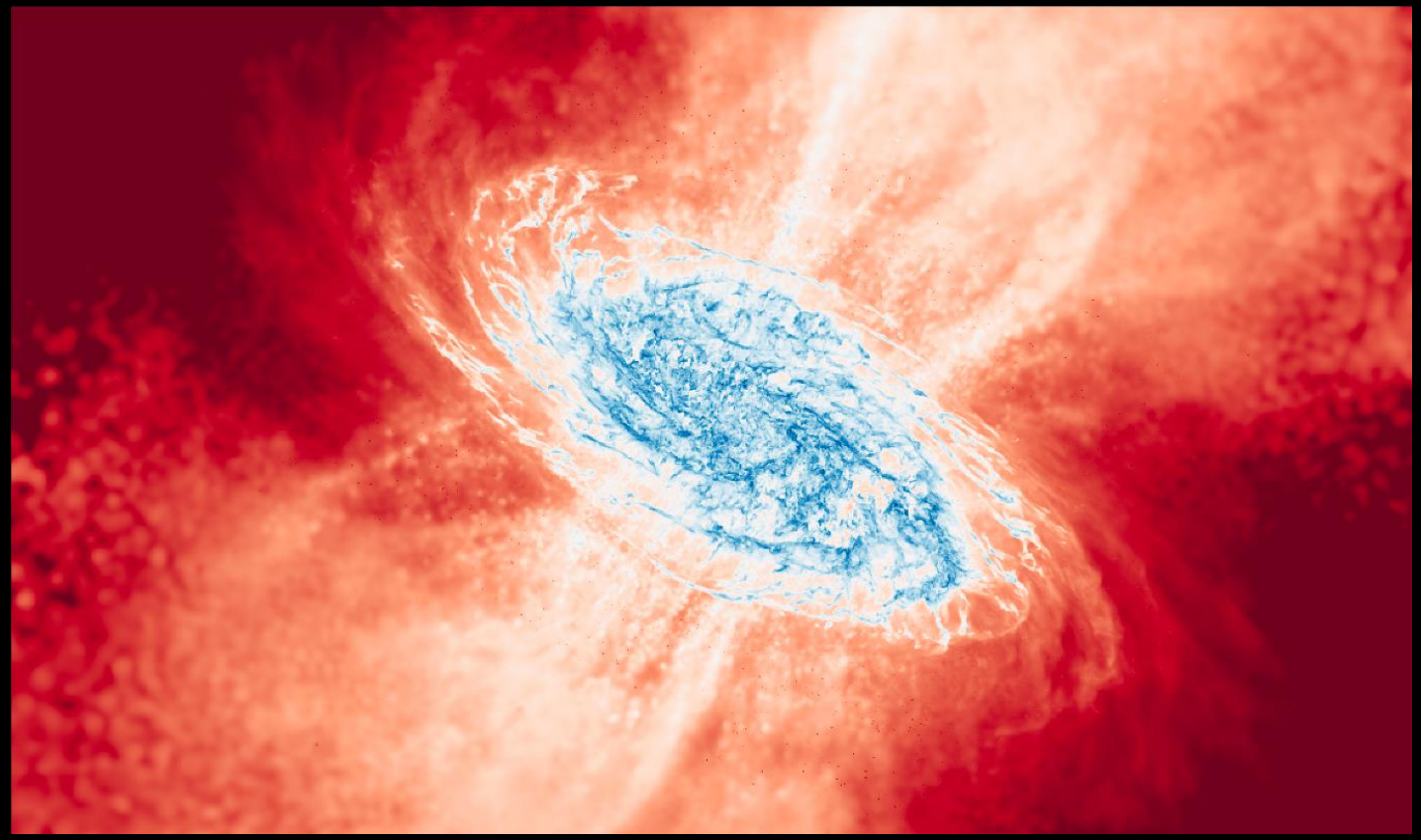
- Correlation between P_{vel} (P_{turb}) and P_{mag}
- Both model see relatively large B-field far from the disc plane: advection, diffusion, or "local" dynamo in the CGM (probably not resolved..)?

Variation on feedback strength

- N_{FB} = 64, weaker feedback generate less outflows, but more "fountain" motions;
- Higher feedback energy increase B field in the CGM as expected;
- Stronger feedback also leads to larger extent of B field in the disc – consistent with $\alpha\omega$ dynamo theory (e.g., Klein 1987)



Conclusions



Main dynamo process:
Strong mean-field dynamo, consistent with the classic theory of an $\alpha\omega$ dynamo.

No Feedback:
GI sets in the disc and the dynamo is collapse driven. The smallest collapsing scale determines amplification

Feedback effect:
Destructive: Increse turbulent diffusion and eject flux into the CGM
Positive increase vertical motions and fountain motions

Resolution dependence:
High resolution dependence; but with our medium resolution the saturation level is largely independent of feedback models;

Numerical dissipation:
Prandtl number below unity
Magnetic Reynolds number in range 10-200
Around the critical value of small-scale dynamo

Initial magnetic field:
No ‘big’ effect in central disk amplification due to early flux ejection to CGM

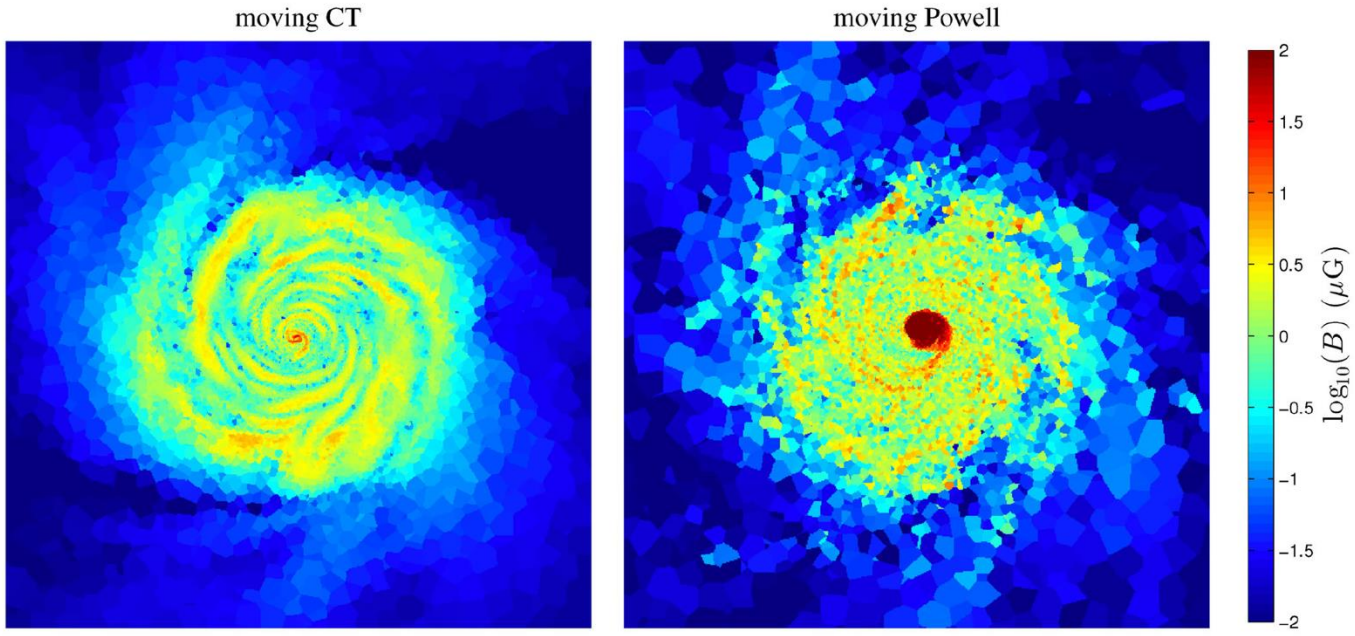


Figure 7. Comparison of the magnetic field strength of the same disc in Fig. 5 at time $t = 0.5$ Gyr in the formation process, simulated using the CT and Powell schemes. The figure displays a physical size of 40 kpc. The CT approach exhibits much better preservation of the topological winding of the magnetic field. The Powell scheme shows substantial divergence error noise seen on the cell level while this is absent to machine precision in CT.

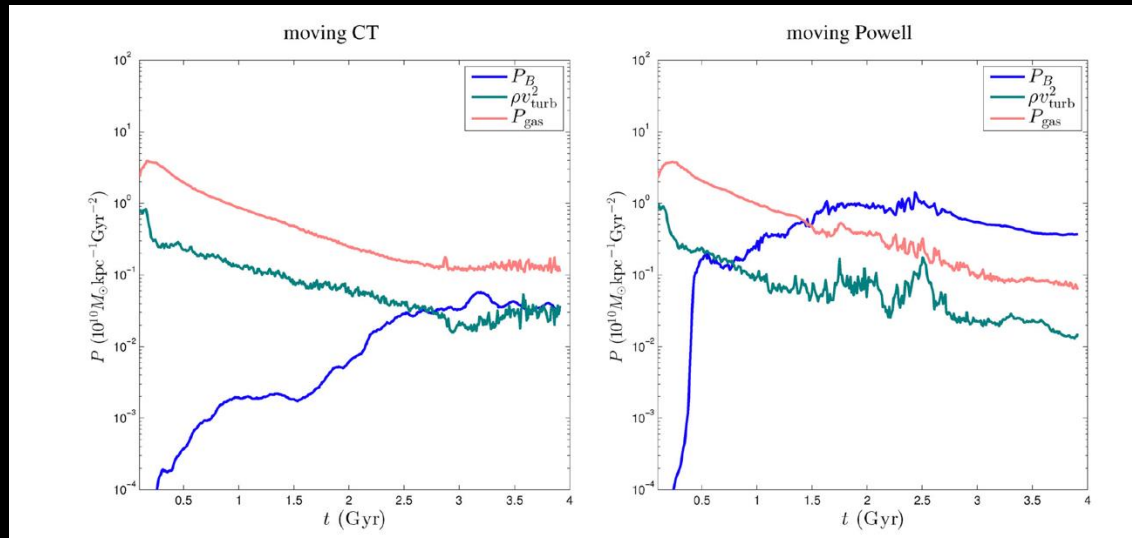
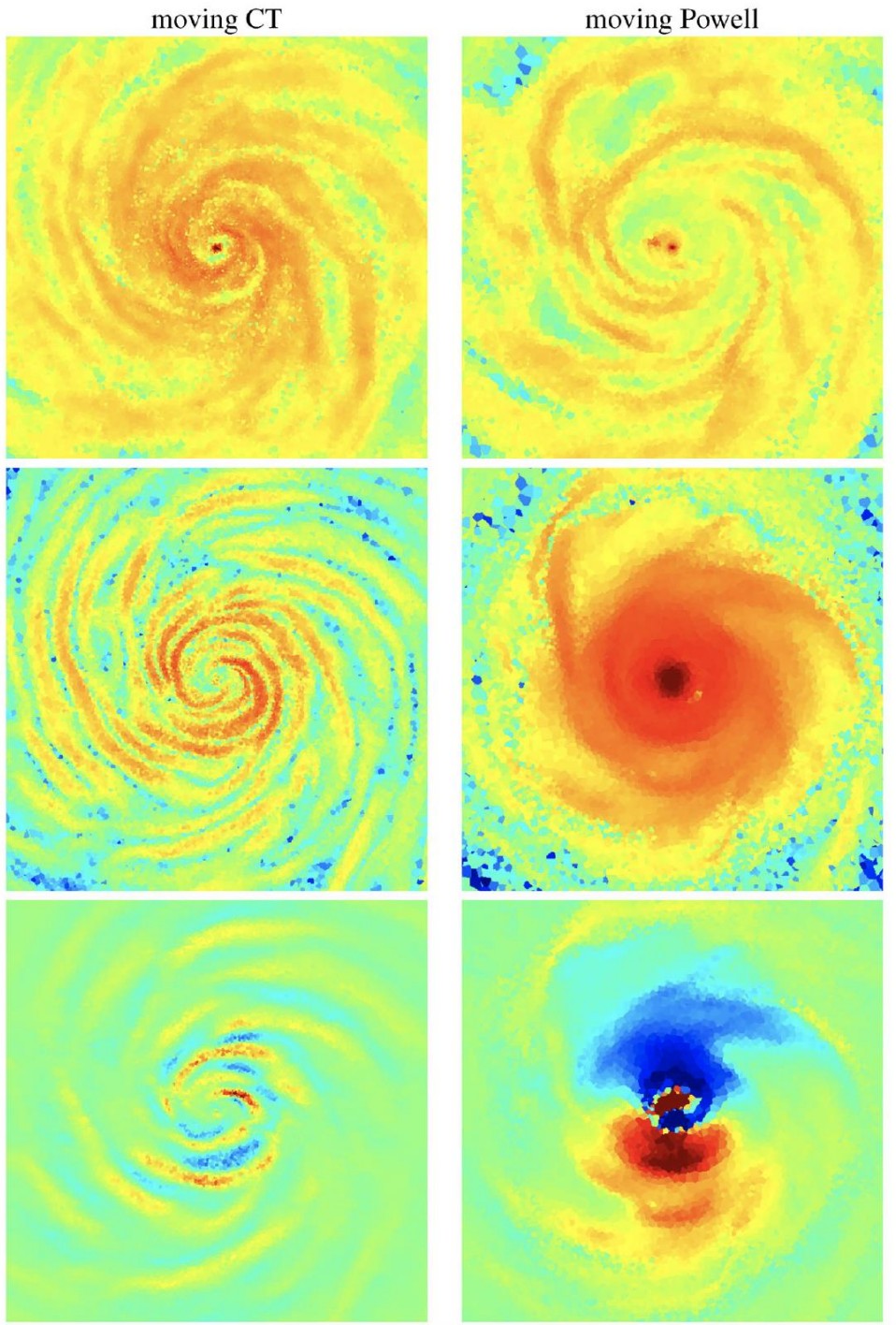


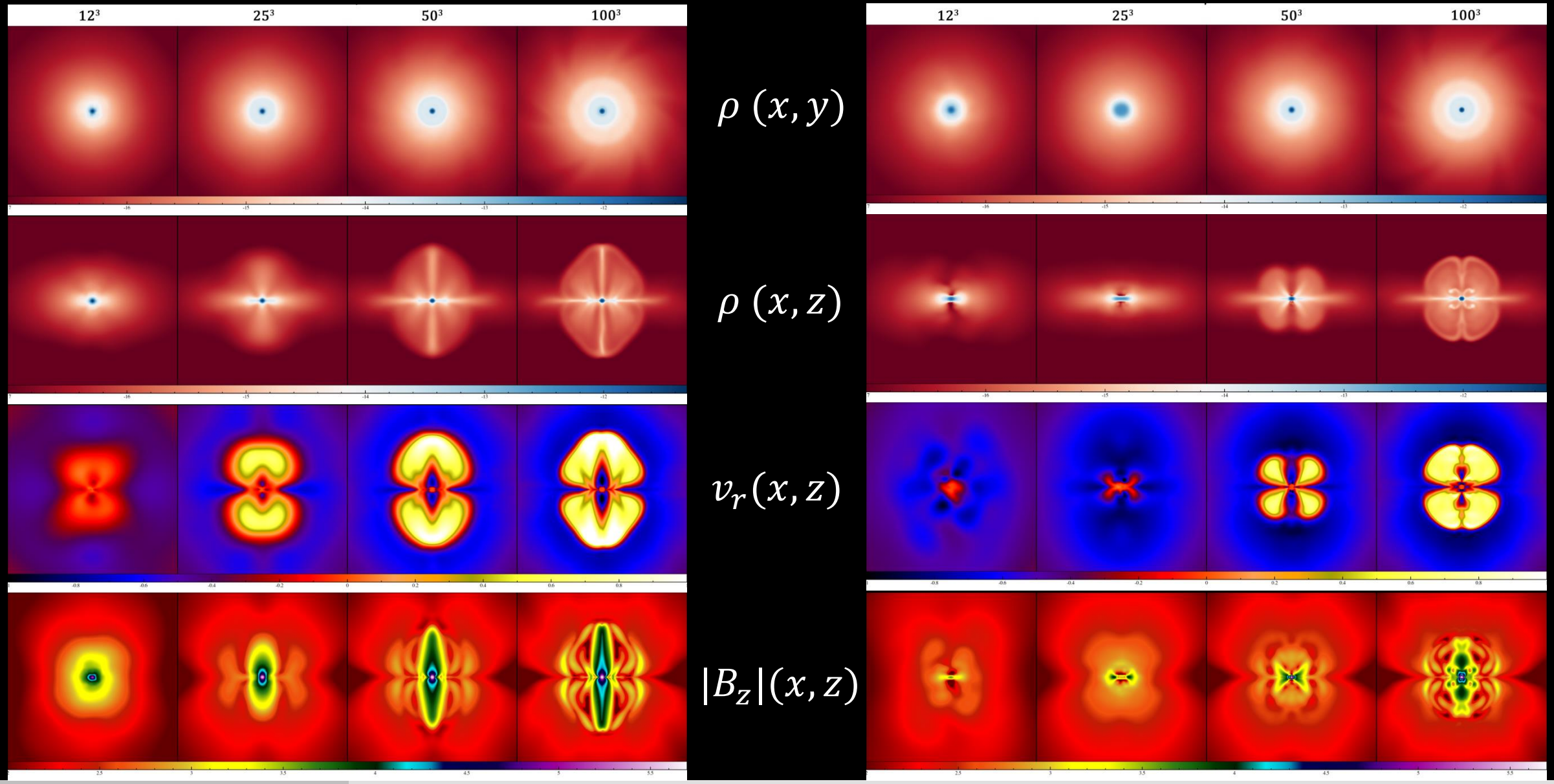
Figure 6. Comparison of magnetic field saturation in the formation of a disc simulated with the CT and Powell schemes. The CT method shows equipartition between magnetic energy density and turbulent kinetic energy density, whereas the Powell technique saturates the field at higher values, exceeding the thermal pressure by about a factor of 5.



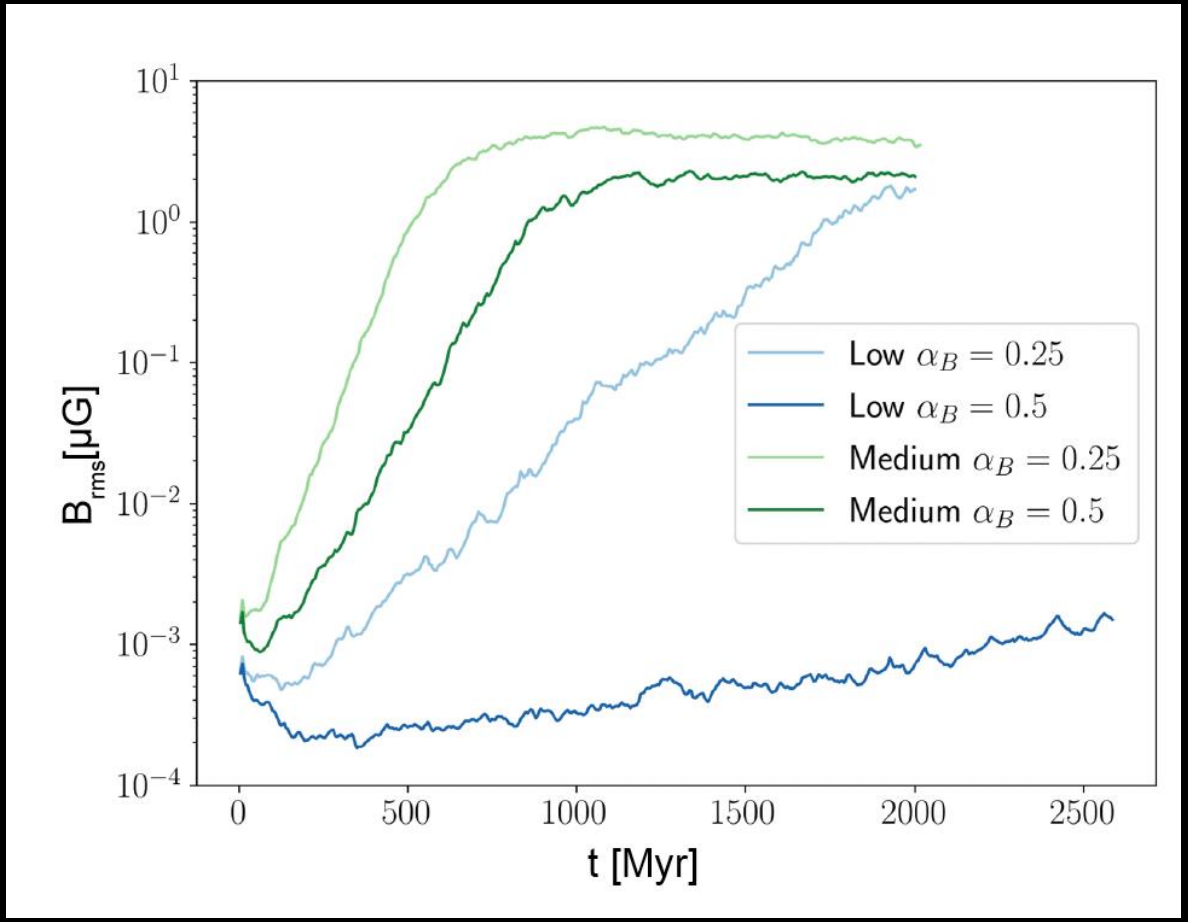
Resolution study

GDSPH

TSPH

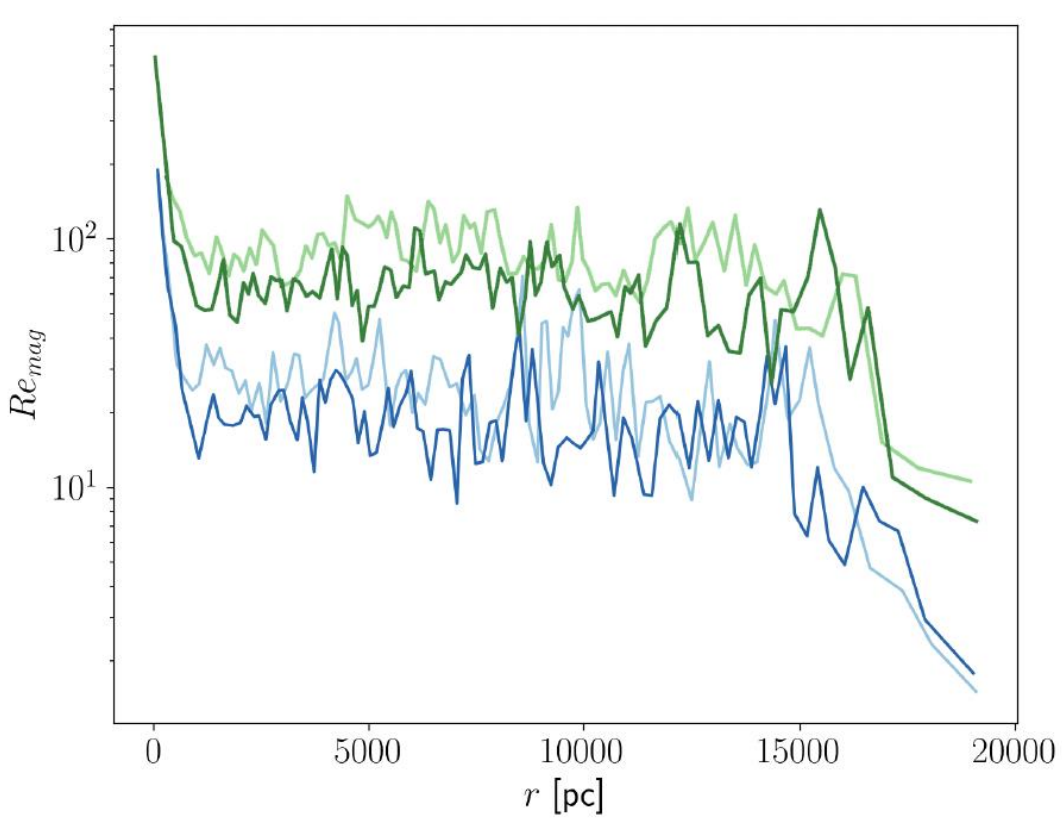


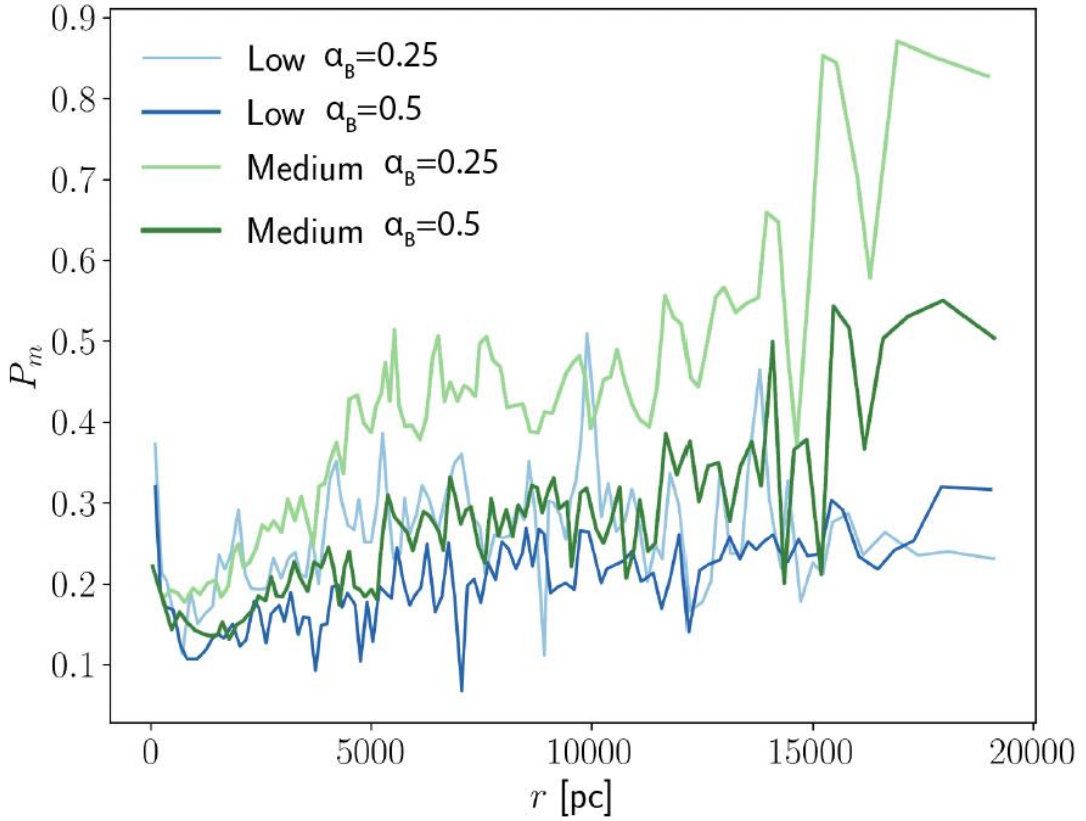
Effect of diffusion parameters



- Lower resistivity generally leads to faster growth and higher saturation level, especially in low resolution cases
- Medium resolution amplification behaviour is more converged

Fluid parameters





- Assuming injection length of 1 kpc, the Rem for medium resolution generally around few 10s to 100s
- Pandtl number remains very low for most of runs in comparison with subsonic, more idealised simulations